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Benefit-Risk Analysis of OX40L Blockade in Hidradenitis Suppurativa

Implementation Details

June 2025

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Document Map

This report is structured to support a wide audience, including executives, scientists, regulatory specialists, and operational teams. For efficient navigation and to quickly find the most relevant sections, please refer to the guide below:

- Executive Summary: A high-level overview of objectives, project steps, outcomes, and strategic importance. Recommended starting point for all readers.
- Section I – Introduction: Project rationale, context, and primary goals.
- Section II – Technical Description and AI Models: Detailed pipeline architecture, modeling steps, and model cards. Recommended for technical, scientific, and implementation teams seeking comprehensive technical knowledge.
- Section III – Quality Process Description (For scientific, quality, and regulatory teams): Covers model validation design and results, dataset construction, risk assessments, limitations, performance metrics, and regulatory alignment. Important for those interested in experimental rigor, benchmark comparisons, or quality assurance.
- Section IV – Technology Deployment Outcomes: For business, operational, and IT leaders. Discusses platform deployment, business value, ROI, real-world use cases, and roadmap for organization-wide impact.

Executive summary

This document provides a comprehensive overview of ArcaScience's AI-driven Benefit-Risk Assessment (BRA) framework, primarily conducted with Sanofi and targeted at Hidradenitis Suppurativa (HS). The project's central question—whether AI can reliably assess the risk-benefit profile of drugs in early preclinical stages to support informed decision-making—was addressed through a meticulously developed multi-step AI pipeline.

Key Project Steps (by order):

- Project Context and Objective:

The primary aim of this project was to assess the feasibility of applying artificial intelligence to systematically extract and compare the risks and benefits of drugs under development—based on their mechanisms of action—from public scientific literature for specific diseases. The project sought not only to evaluate investigational drugs, such as amlitelimab, but also to benchmark these findings against already marketed therapies—in this case, adalimumab for Hidradenitis Suppurativa.

- Technical Pipeline Development:

An advanced pipeline comprising disease modeling, risk extraction, benefit extraction, and data enrichment was created.

Multiple proprietary and custom-developed AI components were used, including sentence classification, named entity recognition, relation extraction, and ontology-driven standardization.

- Model Validation and Performance:

The AI pipeline's individual components were validated both internally and using external datasets, with F1-scores consistently above 85% for extraction tasks.

External benchmarks (e.g., comparison to DisGeNET and other research papers) confirmed the robustness for top-ranked disease-gene associations.

- Quality Assurance and Regulatory Alignment:

The pipeline and its outputs were evaluated and aligned with the FDA's 2025 guidance for AI in regulatory decision-making.

A detailed risk assessment mapped model influence and potential consequence for four key application areas (early target identification, safety signal detection, benefit assessment, regulatory submission), classifying risks as low-medium or medium, and emphasizing the need for human expert oversight throughout.

- Deployment Strategy and Business Impact:

The report proposes transitioning from one-off projects to an integrated, dynamic organization-wide platform at Sanofi.

The platform will offer customizable dashboards, robust data integration, real-time updates, and proactive risk monitoring, delivering up to 71% time savings and €100,000+ in yearly cost reductions, with a projected ROI of 27%.

Use cases span target validation, regulatory submission, and pharmacovigilance, with further future expansions (e.g., social media signal mining).

- Why Implementation Is Important:

Strategic value & transformation: The project transitions BRA from manual, disjointed tasks to a scalable, continuous capability, dramatically accelerating literature reviews, facilitating earlier risk detection, and supporting regulatory and business imperatives in the pharmaceutical value chain.

Cost & time efficiency: Significant reductions in project timelines and outsourcing costs directly enhance the organization's competitive edge and operational agility.

Continuous improvement: Built-in monitoring, user feedback loops, and periodic model retraining ensure the system evolves with both user needs and the scientific landscape.

List Of Abbreviations

Abbreviation	Full Term/Definition
AI	Artificial Intelligence
API	Application Programming Interface
AE	Adverse Event
AS	ArcaScience
BIO	Beginning–Inside–Outside (tagging scheme for NER)
BRA	Benefit-Risk Assessment
COA	Clinical Outcome Assessment
CRO	Contract Research Organization
DLQI	Dermatology Life Quality Index
FDA	Food and Drug Administration
HS	Hidradenitis Suppurativa
IB	Investigator’s Brochure
ISMS	Information Security Management System
IT	Information Technology
LLM	Large Language Model
MAA	Marketing Authorization Application
M2R	Master 2 Recherche (French postgraduate research degree)

MeSH	Medical Subject Headings
MedDRA	Medical Dictionary for Regulatory Activities
NER	Named Entity Recognition
NDA	New Drug Application
OX40L	OX40 Ligand (a molecular target)
PICOS	Population, Intervention, Comparator, Outcomes, Study Design
PMC	PubMed Central
QA	Quality Assurance
R&D	Research and Development
RWE	Real-World Evidence
RE	Relation Extraction
ROI	Return on Investment
SaaS	Software as a Service
SOC	System Organ Class
TPP	Target Product Profile
UAT	User Acceptance Testing
VILT	Virtual Instructor-Led Training

I. Introduction : Project Context

This document serves as a comprehensive report on the implementation project conducted by ArcaScience over the past six months. It presents the technical components, AI models, quality assurance processes, use cases, deliverables, and the planned steps for broader deployment.

The project aimed to evaluate the feasibility of using Artificial Intelligence to automatically extract and identify, from publicly available literature, the potential and actual risks and benefits of preclinical drugs for specific diseases.

At the heart of this initiative was a key question:

Can we leverage AI to assess the risk-benefit profile of a drug—even at the early preclinical stage—to support informed decision-making and accelerate research?

As part of this implementation effort, the team focused on **Hidradenitis Suppurativa (HS)**, a rare disease previously analyzed by ArcaScience. Two drugs were evaluated:

- **Amltelimab**, which targets OX40L and is currently under investigation by Sanofi
- **Adalimumab**, the current standard of care for HS

ArcaScience's technical teams, supported by Sanofi's expert insights, deployed proprietary technologies to process scientific data and deliver key outputs, including:

- A **risk-benefit tree** summarizing insights by System Organ Class (SOC)
- A **synthesis of benefit-risk analyses**
- **Linked data** referencing all source materials for each finding
- A **detailed map of HS pathophysiology**, highlighting relevant molecular targets and key biological pathways, along with their activation levels

This document provides an in-depth overview of the project's technical foundations and quality processes, as well as potential pathways for operational deployment of the solution.

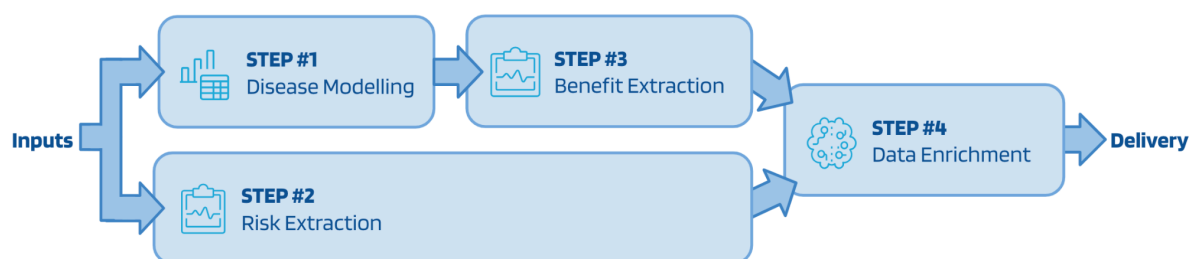
II. Technical Description and AI Models

II.1. Technical Description

The pipeline created for the BRA project is a structured, multi-step process designed to perform a benefit-risk assessment, primarily using our off-the-shelf AI components, supplemented by a few custom-developed modules tailored to this project.

It begins with the collection of data on disease pathophysiology (Step 1). The output from this step feeds into the benefit extraction process (Step 3), alongside additional data sources. In parallel, risk extraction is performed using the name of the intervention or its molecular target as input (Step 2). Finally, the combined information from all previous steps is enriched and synthesized (Step 4) before being delivered as the final output.

BRA Global Pipeline



II.1.1. Step #1: Disease Modelling

- **Inputs:** Pathology Name. This refers to the disease of interest, such as "Hidradenitis Suppurativa".
- **Processing:** This step focuses on the detection of any relations between diseases and any protein or biomarker or gene The process involves several sub-steps:
 - 1.A Sentence classifier model processes the Pathology Name.
 - 2.Standardization of relation is performed using an LLM.
 - 3.Standardization of Biomarker is carried out via MeSH.
 - 4.Score calculation is then performed. In this step, standardized characteristics of the pathology of interest are extracted, enabling the creation of stratified datasets based on the pathology's attributes using Python.
- **Output:** A complete list of biomarkers, proteins, and genes related to the disease of interest is compiled, along with the type and strength of each relationship. This forms the basis for identifying the key pathophysiological characteristics of the pathology. This output is used as inputs for later steps, specifically Benefit Extraction.

STEP #1: Disease Modelling

Detection of any relations between the diseases and any protein or biomarker



II.1.2 Step #2: Risk Extraction

Inputs: Target name(s) and/or drug name(s)

Processing: This step is designed to extract relevant safety evidence associated with the specified drugs or targets. It follows two distinct data paths—Clinical and Preclinical—based on the type of input (drug vs. target):

- **Clinical Data Path:**

A combination of a classifier model, regular expressions (for Adverse Event [AE] ontology matching), and relation extraction (RE) is used. The extracted AEs are then standardized and classified into System Organ Classes (SOCs) using MedDRA.

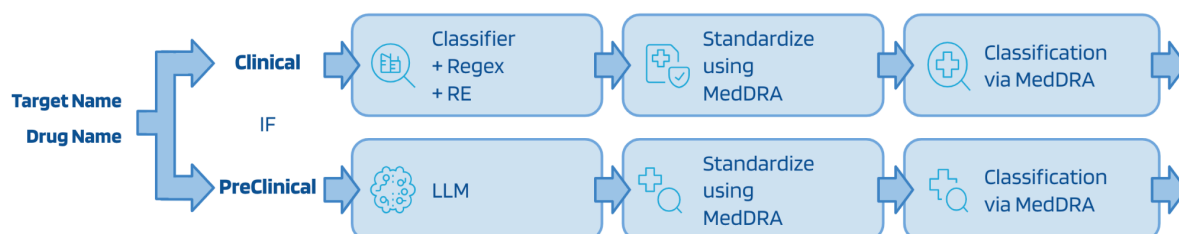
- **Preclinical Data Path:**

A Large Language Model (LLM) is used to extract relevant safety data. The output is then standardized and AEs are classified using MedDRA to ensure consistency with clinical terminology.

Outputs:

A structured dataset containing evidence of drug- or target-associated risks. Each entry includes: the extracted evidence, the associated adverse events, their classification by System Organ Class and the metadata of the source article in which the risk was identified

STEP #2: Risk Extraction



II.1.3 Step #3: Benefit Extraction

Inputs:

Drug or target name, disease name, and biomarkers/targets/proteins (from Step #1)

Processing:

This step is designed to extract relevant benefit evidence associated with the specified drugs or targets within a given disease context. As in the risk extraction step, the process follows two distinct paths: Clinical and Preclinical.

- **Clinical Data Path:**

A combination of models is used, including a classifier model to assess sentence relevance, Regular expressions to detect terms from COA and biomarker ontologies and a relation extraction model to link entities

The output is then processed through a Large Language Model to determine the appropriate System Organ Class classification for each benefit.

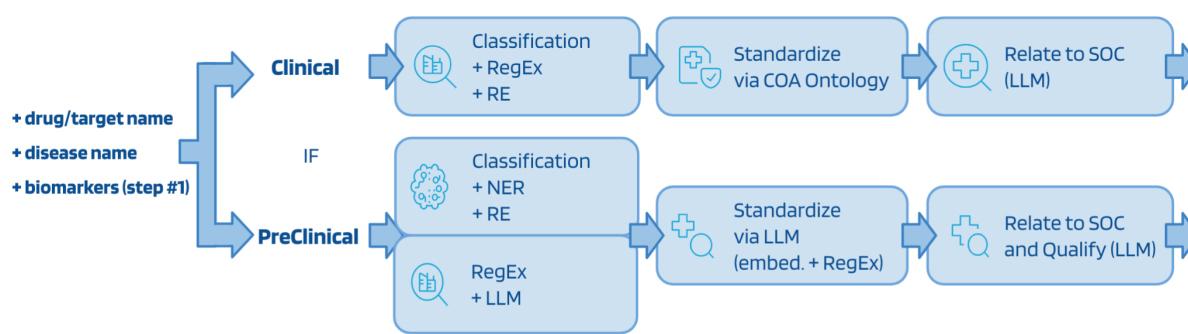
- **Preclinical Data Path:**

The same process used for clinical data is applied. Additionally, a supplementary LLM is deployed to verify the completeness of the extracted information.

Outputs:

A structured dataset containing evidence of drug- or target-associated benefits in the specified indication. Each record includes the extracted benefit evidence, its associated label (e.g., functional improvement, biomarker change), SOC classification, metadata of the source article in which the benefit was identified.

STEP #3: Benefit Extraction



II.1.4 Step #4: Data Enrichment

Inputs: The two benefit and risk tables generated from the previous steps.

Processing: This step focuses on stratification and data contextualization by enriching the benefit and risk information based on details about the Study, Patient population, and Drugs used.

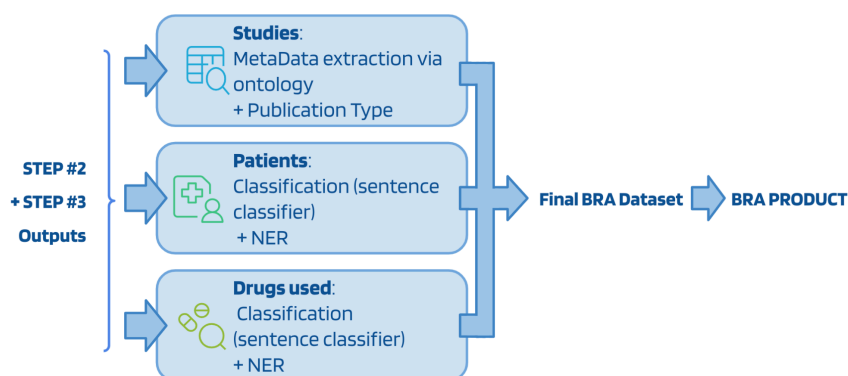
Studies: Metadata is extracted using an ontology and a publication type classification model. For each article, key study attributes are identified, including study design, number and type of arms or groups, randomization approach, blinding method, and study phase.

Patients: Classification using a sentence classifier and NER are applied. Key information extracted includes Age, Sex, Pathology/severity, Race/Ethnicity, Lifestyle Factors, Treatment history, Other Consideration, Species of animals used, and Type of Model. This relates the risk or benefit to the specific patient population involved.

Drugs used: Information on mode of administration, frequency, dosage, and co-administered drugs is extracted through a combination of sentence classification and Named Entity Recognition (NER). Standardization and refinement are supported by NLP techniques, relation extraction prompts, and regular expressions.

Outputs: Contextualized and stratified benefit and risk information. This involves adding columns related to study details (design, phase, randomization, blinding) and patient characteristics (age, sex, condition, sample size) to the benefit and risk dataset. The final deliverables for benefits and risks are then available for analysis.

STEP #4: Data Enrichment



II.2. BRA Models

This section details the Artificial Intelligence models and pipelines developed in-house for this project, outlining their architecture and functionality.

II.2.1. Data Enrichment Pipeline and Model Cards

Our data enrichment pipeline enhances extracted benefit and risk information with critical contextual metadata, enabling more nuanced analysis and interpretation of findings. First, we extract study metadata from our pre-annotated articles & data, leveraging publication type classifications already assigned by our publication type classifier. This provides immediate context regarding the evidence quality and study methodology. We then apply targeted regex patterns to identify key study characteristics, including design type (cohort, in vivo, open label, etc.), arm groups (SOC, placebo, combination therapy), randomization

status, blinding methodology, and study phase. These structured attributes allow for stratification of evidence by methodological rigor and study context.

The patient enrichment component utilizes a dedicated classification model to identify sentences containing patient-related information, following our standard three-class approach (relevant, possibly relevant, irrelevant). Sentences classified as containing patient information are processed by a specialized NER model that extracts detailed patient characteristics such as demographics, inclusion/exclusion criteria, baseline characteristics, and treatment history. This extraction enables correlation between patient populations and observed benefit patterns or safety signals, critical for understanding potential subgroup effects and generalizability of findings. The patient information extraction achieves strong performance particularly on well-structured clinical trial reports and patient characterization sections.

Drug mentions and their administration details are systematically extracted through our Drug Modality NER model. This specialized NER component not only identifies explicit and implicit drug references across various naming conventions (generic, brand, chemical, abbreviated) but also extracts critical modality information, including dosage, frequency, duration, and mode of administration. This comprehensive extraction enables detailed therapeutic regimen analysis, allowing correlation between specific treatment protocols and observed outcomes. The model's multi-entity approach captures the full context of drug administration, providing insights into dose-response relationships, administration route influences, and treatment duration effects. The extracted drug entities are subsequently standardized using our drug ontology to ensure consistent identification across terminology variations.

II.2.1.1. Model Card: Patient Classification Model (PCM)

Patient Classification Model - Model Card			
Model Version:	1.2.0	Last Updated:	November 29, 2024
Type:	Encoder-based Text Classification	Language(s):	English
Framework:	PyTorch		

Patient Classification Model - Model Description
Overview
This model identifies sentences in biomedical articles that contain patient-related information. It serves as the screening component in our patient data extraction pipeline, selecting sentences that describe patient demographics, characteristics, inclusion/exclusion criteria, or treatment history. The model analyzes sentences and

classifies them into three categories based on their relevance to patient information, enabling targeted extraction of patient data for downstream analysis.

Intended Use

This model is intended to filter biomedical text to identify sentences containing patient information, which can then be processed by the Patient NER model for detailed extraction. The three-class approach allows for capturing clearly relevant patient descriptions while also retaining potentially useful sentences with implicit patient information. It is designed for use in clinical evidence synthesis, patient population analysis, and subgroup identification.

Out-of-Scope Use

This model is not intended for:

- Processing non-biomedical text
- Complete patient data extraction without the downstream NER component
- Diagnosis or individual patient identification

Patient Classification Model - Model Architecture

Base Architecture:	pretrained biomedical model	Number of Parameters:	~110M
Input Format:	Tokenized text (maximum 512 tokens)	Output Format:	Three-class classification (0: Irrelevant, 1: Possibly relevant, 2: Relevant)
Key Architectural Features:			
→ Fine-tuned high-end healthtech encoder large model with classification head → Linear layer (768 → 3) with softmax activation → Pretrained on biomedical corpus for domain-specific language understanding			

Patient Classification Model - Training Details

Training Data

- Dataset(s): Proprietary annotated biomedical articles
- Size: 34,820 annotated sentences
- Data Format: Sentences extracted from biomedical articles with three-class labels
- Preprocessing: Sentence segmentation, direct tokenization using high-end healthtech encoder tokenizer
- Data Split: 70% training, 15% validation, 15% test
- For Classification:

◆ Classes: Three-class (Relevant / Possibly relevant / Irrelevant) ◆ Class Distribution: 25% relevant, 35% possibly relevant, 40% irrelevant
Training Procedure
→ Optimizer: AdamW → Learning Rate: 2e-5 with linear decay → Batch Size: 32 → Sequence Length: 512 → Training Steps/Epochs: 4 epochs → Loss Function: Cross-entropy → Regularization: Dropout (0.1) → Early Stopping Criteria: No improvement in validation macro F1 score for 3 consecutive epochs
Evaluation Metrics
→ Primary Metrics: Macro F1 score, Per-class Precision, Recall → Secondary Metrics: Accuracy → Evaluation Method: Hold-out test set (15% of data)

Patient Classification Model - Performance Results - Overall Performance:		
Metric	Value	Notes
Macro Precision	86.4%	Averaged across the three classes
Macro Recall	82.7%	Averaged across the three classes
Macro F1 Score	84.5%	Balanced measure across all classes
Accuracy	87.2%	Overall correctness

Patient Classification Model - Performance Results - Error Analysis:	
Common error patterns include:	
→ Lower recall for implicit patient information	

Patient Classification Model - Limitations and Considerations:
Known Limitations → Performance varies based on how explicitly the patient information is stated

Patient Classification Model - Maintenance and Support:
Monitoring and Maintenance → Monitoring Plan: Evaluation on sample of new literature every 2 months → Retraining Criteria: >3% decrease in F1 score on monitoring set
Versioning → Current Version: 1.2.0 → Update Frequency: Quarterly evaluation, annual retraining

II.2.1.2. Model Card: Patient NER Model

Patient NER Model - Model Card			
Model Version:	1.2.0	Last Updated:	November 29, 2024
Type:	Encoder-based Named Entity Recognition	Language(s):	English
Framework:	PyTorch		

Patient NER Model - Model Description
Overview
This model extracts detailed patient information from sentences in biomedical articles, identifying spans of text that describe specific patient characteristics. Following initial sentence classification by the Patient Classification Model, this NER component identifies and categorizes patient-related entities such as demographics (age, gender, ethnicity), eligibility criteria, baseline clinical characteristics, and treatment history. These structured data points enable patient population analysis and correlation of findings with specific patient subgroups.

Intended Use
This model is intended for extracting structured patient information from biomedical text to enable population characterization, subgroup analysis, and assessment of study generalizability. It specifically targets clinical trial reports, observational studies, and other literature with patient descriptions.
Out-of-Scope Use
This model is not intended for: → Processing non-biomedical text → Identifying individual patients → De-anonymizing study participants

Patient NER Model - Model Architecture			
Base Architecture:	pretrained biomedical model	Number of Parameters:	~110M
Input Format:	Tokenized text (maximum 256 tokens)	Output Format:	Token-level BIO tagging for patient-related entities
Key Architectural Features:			
→ Fine-tuned high-end healthtech encoder large model for token classification → Linear layer for token classification (768 → number of entity types × 3) → BIO tagging scheme for entity boundary detection			

Patient NER Model- Training Details	
Training Data	
→ Dataset(s): Proprietary annotated patient descriptions from biomedical articles → Size: 29,790 sentences with token-level annotations → Data Format: Sentences with BIO tags for patient-related entities → Preprocessing: Sentence segmentation, direct tokenization using high-end healthtech encoder tokenizer → Data Split: ◆ 70% training, ◆ 15% validation, ◆ 15% test → For Named Entity Recognition: ◆ Entity Types: Demographics, Inclusion Criteria, Exclusion Criteria, Baseline Characteristics, Treatment History, Comorbidities ◆ Annotation Guidelines: Multi-phase expert annotation with adjudication by clinical professionals	

Training Procedure
→ Optimizer: AdamW → Learning Rate: 3e-5 with linear decay → Batch Size: 16 → Sequence Length: 256 → Training Steps/Epochs: 5 epochs → Loss Function: Token-level cross-entropy → Regularization: Dropout (0.1) → Early Stopping Criteria: No improvement in validation entity F1 score for 3 consecutive epochs
Evaluation Metrics
→ Primary Metrics: Entity-level F1 score, Precision, Recall → Secondary Metrics: Token-level accuracy, Span detection accuracy → Evaluation Method: Hold-out test set (15% of data)

Patient NER Model - Performance Results - Overall Performance:		
Metric	Value	Notes
Entity Precision	87.8%	Exact match at entity level
Entity Recall	83.5%	Exact match at entity level
Entity F1 Score	85.6%	Harmonic mean of precision and recall
Token Accuracy	94.3%	Token-level classification accuracy

Patient NER Model - Performance Results - Error Analysis:
Common error patterns include: → Lower recall for rare or uncommon patient characteristics → Possible confusion between inclusion and exclusion criteria

Patient NER Model - Limitations and Considerations:
Known Limitations
→ Entity extraction performance depends on sentence-level classification quality → Performance varies across publication types and study designs

Patient NER Model - Maintenance and Support:	
Monitoring and Maintenance	
→ Monitoring Plan: Evaluation on sample of new literature every 2 months	
→ Retraining Criteria: >3% decrease in F1 score on monitoring set	
Versioning	
→ Current Version: 1.2.0	
→ Update Frequency: Quarterly evaluation, annual retraining	

II.2.1.3. Model Card: Drug Modality NER Model

Drug Modality NER Model - Model Card			
Model Version:	1.2.0	Last Updated:	December 05, 2024
Type:	Encoder-based Named Entity Recognition	Language(s):	English
Framework:	PyTorch		

Drug Modality NER Model - Model Description	
Overview	
This model identifies and extracts drug mentions and their administration details from biomedical text. It recognizes various forms of drug references including generic names, brand names, chemical descriptors, and abbreviations, while simultaneously extracting critical prescription and administration information such as dosage, frequency, duration, and mode of administration. It's supplementary to our drug ontology. This comprehensive extraction creates a detailed profile of therapeutic regimens discussed in the literature, enabling analysis of treatment protocols and their relationship to clinical outcomes.	
Intended Use	
This model is intended for comprehensive drug entity extraction from biomedical literature to support therapeutic landscape analysis, comparative effectiveness research, and drug-biomarker relationship mapping. It complements the benefit and risk detection pipelines by providing pharmaceutical context to identified biological relationships.	

Out-of-Scope Use

This model is not intended for:

- Processing non-biomedical text
- Stockage and conservation information

Drug Modality NER Model - Model Architecture

Base Architecture:	pretrained biomedical model	Number of Parameters:	~110M
Input Format:	Tokenized text (maximum 256 tokens)	Output Format:	Token-level BIO tagging for drug entities
Key Architectural Features:			
→ Fine-tuned high-end healthtech encoder large model for token classification → Linear layer for token classification → BIO tagging scheme for entity boundary detection			

Drug Modality NER Model - Training Details**Training Data**

- Dataset(s): Proprietary annotated drug mentions from biomedical articles
- Size: 42,182 sentences with token-level annotations
- Data Format: Sentences with BIO tags for drug entities
- Preprocessing: Sentence segmentation, direct tokenization using high-end healthtech encoder tokenizer
- Data Split:
 - ◆ 70% training,
 - ◆ 15% validation,
 - ◆ 15% test
- For Named Entity Recognition:
 - ◆ Entity Types: Drug, Dosage, Frequency, Duration, Mode of Administration
 - ◆ Annotation Guidelines: Expert annotation with pharmaceutical reference validation

Training Procedure

- Optimizer: AdamW
- Learning Rate: 3e-5 with linear decay
- Batch Size: 32
- Sequence Length: 512

- Training Steps/EPOCHS: 4 epochs
- Loss Function: Token-level cross-entropy
- Regularization: Dropout (0.1)
- Early Stopping Criteria: No improvement in validation entity F1 score for 3 consecutive epochs

Evaluation Metrics

- Primary Metrics: Entity-level F1 score, Precision, Recall
- Secondary Metrics: Token-level accuracy, Span detection accuracy
- Evaluation Method: Hold-out test set (15% of data)

Drug Modality NER Model - Performance Results - Overall Performance:

Metric	Value	Notes
Entity Precision	92.5%	Exact match at entity level
Entity Recall	89.7%	Exact match at entity level
Entity F1 Score	91.1%	Harmonic mean of precision and recall
Token Accuracy	96.8%	Token-level classification accuracy

Drug Modality NER Model - Performance Results - Error Analysis:

Common error patterns include:

- Occasional false positives with chemical compounds not used therapeutically

Drug Modality NER Model - Limitations and Considerations:**Known Limitations**

- Limited ability to discern drug roles (target therapy vs. comparator vs. concomitant)

Drug Modality NER Model - Maintenance and Support:	
Monitoring and Maintenance	
→ Monitoring Plan: Monthly evaluation on sample of new literature → Retraining Criteria: >3% decrease in F1 score on monitoring set	
Versioning	
→ Current Version: 2.1.0 → Update Frequency: Quarterly evaluation, annual retraining	

II.2.1.4. Model Card: Publication Type Classification Model

Publication Type Classification Model - Model Card			
Model Version:	1.0.0	Last Updated:	October 25, 2024
Type:	Document-level Classification	Language(s):	English
Framework:	PyTorch	Created by:	Data Enrichment Team

Publication Type Classification Model - Model Description	
Overview	
This model classifies biomedical documents into specific publication types, identifying the research methodology and article category. It analyzes document structure, content, and linguistic patterns to categorize articles into 10 types: clinical trials, observational studies, studies, reviews, meta-analyses, systematic reviews, case reports, abstracts, guidelines/protocols, and other. This classification provides critical context for interpreting our findings, as the evidentiary value of results varies substantially by publication type.	
Intended Use	
This model is intended for automatically categorizing biomedical literature by publication type to enable evidence stratification, quality assessment, and appropriate weighting of findings in evidence synthesis.	
Out-of-Scope Use	
This model is not intended for: → Content-level quality assessment → Publication acceptance decisions	

Publication Type Classification Model - Model Architecture			
Base Architecture:	pretrained biomedical model	Number of Parameters:	~110M
Input Format:	Tokenized text (maximum 256 tokens)	Output Format:	Multi-class classification across publication types
Key Architectural Features:			
→ Fine-tuned high-end healthtech encoder large model as encoder → Bi-directional LSTM head for improved sequence modeling → Special tokens are used for boundary markers for document structure → Only processes the most relevant content of an article to handle document length constraints			

Publication Type Classification Model - Training Details	
Training Data	
→ Dataset(s): Labeled biomedical articles and public PubMed metadata → Size: 30,645 documents with publication type labels → Data Format: Structured document representations with publication type labels → Preprocessing: ◆ Document parsing to extract key sections ◆ Selection of high-information content: title, abstract, figure/table captions, table of contents, and strategically sampled full text ◆ Format optimization to fit within 512 token limit while preserving critical indicators of publication type → Data Split: ◆ 70% training, ◆ 15% validation, ◆ 15% test. → For Classification: ◆ Classes: clinical trials, observational studies, studies, reviews, meta-analyses, systematic reviews, case reports, abstracts, guidelines/protocols, and other ◆ Class Distribution: Balanced through stratified sampling	
Training Procedure	
→ Optimizer: AdamW → Learning Rate: 1e-5 with linear decay → Batch Size: 8	

→ Sequence Length: 512 → Training Steps/EPOCHS: 3 epochs → Loss Function: Cross-entropy → Regularization: Dropout (0.1) → Early Stopping Criteria: No improvement in validation accuracy for 2 consecutive epochs
Evaluation Metrics
→ Primary Metrics: Accuracy, Macro F1 score, Per-class Precision, Recall → Secondary Metrics: Confusion matrix → Evaluation Method: Hold-out test set (15% of data)

Publication Type Classification Model - Performance Results - Overall Performance:		
Metric	Value	Notes
Accuracy	83.79%	Overall classification correctness
Macro Precision	83.66%	Averaged across publication types
Macro Recall	86.25%	Averaged across publication types
Macro F1 Score	83.85%	Balanced measure across all types

Publication Type Classification Model - Performance Results - Error Analysis:
Common error patterns include: → Confusion between closely related publication types (e.g., studies vs. observational studies) → Challenges with mixed-method studies

Publication Type Classification Model - Limitations and Considerations:
Known Limitations
→ Performance depends on document completeness and structure → Limited ability to classify novel or hybrid study designs

Publication Type Classification Model - Maintenance and Support:	
Monitoring and Maintenance	
→ Monitoring Plan: Evaluation on sample of new literature every 2 months → Retraining Criteria: >5% decrease in F1 score on monitoring set	
Versioning	
→ Current Version: 1.0.0 → Update Frequency: Quarterly evaluation, annual retraining	

II.2.2. Disease modelling

Our pipeline begins with an in-house trained sentence classifier that analyzes biomedical text and categorizes sentences as "**Relevant**," "**Possibly Relevant**," or "**Irrelevant**" in the context of relationships between biomarkers and pathologies. Only sentences classified as "Relevant" or "Possibly Relevant" continue through the pipeline for further processing, allowing us to efficiently filter large volumes of text.

These filtered sentences are then fed to a Large Language Model (LLM) that performs extraction and initial standardization. The LLM identifies the specific biomarkers mentioned, their locations within the body, and the type of relationship they have with the pathology (such as "Upregulated," "Activated," or "Downregulated"). During this step, we also standardize biomarker names according to our curated ontology, ensuring consistent terminology throughout the analysis.

Following extraction, our **first post-processing** step involves location stratification using a rule-based approach. We categorize the extracted locations into three groups: Systemic (blood or serum), Local (specific to the pathology, such as hair follicles or HS lesions), and Other (any location not fitting the previous categories). For each biomarker and location combination, we count the number of supporting articles and calculate a statistical consensus score that reflects the level of agreement across sources regarding the expression alteration.

The final **post-processing step** identifies meaningful pathology characteristics by applying specific filtering criteria to the stratified data. We retain only those biomarker-pathology relationships that are not classified as "Non-altered," are mentioned in more than three articles, and have a statistical consensus of at least 75% in at least one location type. This filtering ensures that we focus only on the most significant and well-supported biomarker relationships, providing valuable insights into the pathological mechanisms of the disease under study.

II.2.2.1. Model Card: Biomarker-Pathology Relation Sentence Classifier

Biomarker-Pathology Relation Sentence Classifier - Model Card			
Model Version:	1.0.0	Last Updated:	October 25, 2024
Type:	Document-level Classification	Language(s):	English
Framework:	PyTorch	Created by:	Data Enrichment Team

Biomarker-Pathology Relation Sentence Classifier - Model Description
Overview
This model identifies sentences in biomedical articles that contain relations between pathologies and biomarkers. It serves as the first component in a pipeline for extracting standardized biomarker-pathology relationships from scientific literature. The model analyzes sentences and classifies them into three categories based on their relevance to biomarker-pathology relationships, enabling more nuanced filtering of content for downstream processing. These 3 categories are: → relevant, → possibly relevant, → irrelevant.
Intended Use
This model is intended to filter large volumes of biomedical text to identify only those sentences that contain biomarker-pathology relationships, which can then be processed by downstream models for entity extraction and relation classification. It is specifically designed for use in biomedical research and clinical evidence synthesis for understanding disease mechanisms.
Out-of-Scope Use
This model is not intended for: → Direct clinical decision-making → Processing non-biomedical text → Identifying other types of biological relationships not involving protein, biomarker, gene, sRNA, hormone, cell, immune markers, and pathologies → Complete relationship extraction without the downstream components

Biomarker-Pathology Relation Sentence Classifier - Model Architecture			
Base Architecture:	pretrained biomedical model		Number of Parameters: ~110M
Input Format:	Tokenized text (maximum 512 tokens)	Output Format:	Three-class classification (0: Irrelevant, 1: Possibly relevant, 2: Relevant)
Key Architectural Features:			
→ Fine-tuned high-end healthtech encoder large model with a classification head → Linear layer (768 → 3) with softmax activation → Pretrained on a biomedical corpus for domain-specific language understanding			

Biomarker-Pathology Relation Sentence Classifier - Training Details	
Training Data	
→ Dataset(s): Publicly available biomedical articles → Size: 11,252 → Data Format: Sentences extracted from biomedical articles with three-class labels → Preprocessing: ◆ Sentence segmentation, ◆ direct tokenization using high-end healthtech encoder tokenizer → Data Split: ◆ 70% training, ◆ 15% validation, ◆ 15% test → For Classification: ◆ Classes: Three-class (Relevant / Possibly relevant / Irrelevant) ◆ Class Distribution: 25% relevant, 10% possibly relevant, 65% negative	
Training Procedure	
→ Optimizer: AdamW → Learning Rate: 2e-5 with linear decay → Batch Size: 16 → Sequence Length: 512 → Loss Function: cross-entropy → Regularization: Dropout (0.1) → Early Stopping Criteria: No improvement in validation F1 score for 3 consecutive epochs	

Evaluation Metrics

- Primary Metrics: Macro F1 score, Precision, Recall
- Secondary Metrics: Accuracy
- Evaluation Method: Hold-out test set (15% of data)

Biomarker-Pathology Relation Sentence Classifier - Performance Results - Overall Performance:

Metric	Value	Notes
Macro Precision	88.6%	Ability to avoid false positives
Macro Recall	85.8%	Ability to find all relevant sentences
Macro F1 Score	87.1%	Harmonic mean of precision and recall
Accuracy	92.3%	Overall correctness

Biomarker-Pathology Relation Sentence Classifier - Limitations and Considerations:**Known Limitations**

- Performance varies based on how explicitly the relationship is stated

Ethical Considerations and Biases

- Potential biases toward well-studied diseases and biomarkers
- May underrepresent rare diseases due to limited literature

Biomarker-Pathology Relation Sentence Classifier - Maintenance and Support:**Monitoring and Maintenance**

- Monitoring Plan: Monthly evaluation on sample of new literature
- Retraining Criteria: >5% decrease in F1 score on monitoring set

Versioning

- Version History: Initial release

II.2.3. Benefit Pipeline

Our benefit detection pipeline begins with our proprietary efficacy classification model that analyzes biomedical text to identify sentences containing any potential therapeutic benefits. This specialized model categorizes sentences as **"Relevant," "Possibly Relevant,"** or **"Irrelevant"** based on their potential to contain efficacy information. Only sentences classified as "Relevant" or "Possibly Relevant" continue through the pipeline for further analysis, creating an efficient first-pass filter that significantly reduces the volume of text requiring deeper processing.

Following initial classification, we employ a regex-based filtering system to exclude sentences that explicitly mention our drug of interest. This filtering mechanism substantially improves the precision of our downstream analyses by removing potential sources of bias from the dataset.

The filtered sentences then undergo analysis using our comprehensive Clinical Outcome Assessment (COA) ontology to detect and standardize benefit descriptions. This ontology-driven approach identifies specific therapeutic outcomes mentioned in the text, such as "pain reduction," "improved mobility," or "decreased inflammation," and maps them to standardized terms within our framework. This standardization process ensures consistent terminology throughout the analysis and enables quantitative aggregation of benefits reported across multiple sources.

Concurrent with benefit detection, we implement our biomarker ontology to identify any biomarkers mentioned within the context of therapeutic benefits. This specialized ontology recognizes molecular, cellular, and physiological indicators such as "IL-6," "T-cell count," or "cortisol levels" that are associated with treatment response or disease progression. By capturing these biomarker-benefit associations, we gain valuable insights into the potential mechanisms of action and create opportunities for developing more targeted therapeutic approaches.

The final step in our pipeline employs our **unified relation extraction (RE) classification model**, which evaluates whether the identified efficacy information specifically relates to the target pathology. This model analyzes the contextual relationships between efficacy claims and disease mentions to determine relevance, categorizing the results as **"Relevant," "Possibly Relevant,"** or **"Irrelevant"** to the specific pathology under investigation. This critical filtering ensures that only pathology-specific efficacy data influences our subsequent analyses and decision-making processes.

Our PreClinical efficacy evaluation follows a similar workflow with key adaptations to address the unique characteristics of preclinical literature. We begin with the same efficacy classification model, but due to the more complex and variable nature of preclinical reporting, this model alone does not achieve sufficient recall for our requirements. The scientific language in preclinical studies often uses more diverse terminology and experimental frameworks that our classification model wasn't fully optimized to capture.

To address this limitation in the PreClinical pipeline, we supplement the initial classification with a Large Language Model specifically fine-tuned on preclinical literature. This LLM performs a more nuanced analysis of the text, identifying implicit benefits and experimental outcomes that might be described in specialized or technical language. The combination of our classification model and the LLM significantly improves our ability to detect relevant efficacy information in preclinical studies, ensuring comprehensive data capture while maintaining high precision.

II.2.3.1. Model Card: ArcaEfficacy (Efficacy Classification Model)

ArcaEfficacy (Efficacy Classification Model) - Model Card			
Model Version:	1.1.3	Last Updated:	December 05, 2024
Type:	Encoder-based Text Classification	Language(s):	English
Framework:	PyTorch	Created by:	Data Enrichment Team

ArcaEfficacy (Efficacy Classification Model) - Model Description
Overview
This model identifies sentences in biomedical literature that contain information about therapeutic efficacy or benefits. It serves as the initial component in our pipeline for extracting and standardizing efficacy information from scientific publications and clinical trial documentation. The model analyzes sentences and classifies them into three categories based on their relevance to therapeutic benefits: → relevant, → possibly relevant, → and irrelevant.
Intended Use
This model is designed to filter large volumes of biomedical text to identify sentences that describe therapeutic benefits or efficacy outcomes, which are then processed by downstream components for more detailed extraction and standardization. It is specifically intended for use in pharmaceutical research, clinical evidence synthesis, and comparative effectiveness research.
Out-of-Scope Use
This model is not intended for: → Processing non-biomedical text

- Identifying adverse events or safety concerns
- Extracting numerical efficacy data without downstream components
- Evaluating regulatory compliance or marketing claims

ArcaEfficacy (Efficacy Classification Model) - Model Architecture

Base Architecture:	pretrained biomedical model	Number of Parameters:	~110M
Input Format:	Tokenized text (maximum 512 tokens)	Output Format:	Three-class classification (0: Irrelevant, 1: Possibly relevant, 2: Relevant)
Key Architectural Features:			
→ Fine-tuned high-end healthtech encoder large model with a classification head → Linear layer (768 → 3) with softmax activation			

ArcaEfficacy (Efficacy Classification Model) - Training Details

Training Data

- Dataset(s): corpus of clinical trial publications, meta-analyses, systematic reviews, and other publication types
- Size: 15,467 annotated sentences
- Data Format: Sentences extracted from biomedical articles with three-class labels
- Preprocessing: Sentence segmentation
- Data Split:
 - ◆ 70% training,
 - ◆ 15% validation,
 - ◆ 15% test
- For Classification:
 - ◆ Classes: Three-class (Relevant / Possibly relevant / Irrelevant)
 - ◆ Class Distribution: 31.7% relevant, 12.3% possibly relevant, 56.1% irrelevant

Training Procedure

- Optimizer: AdamW
- Learning Rate: 3e-5 with linear decay
- Batch Size: 16
- Sequence Length: 256
- Loss Function: Weighted cross-entropy (weights adjusted for class imbalance)
- Regularization: Dropout (0.1), L2 weight decay (0.01)

→ Early Stopping Criteria: No improvement in validation F1 score for 3 consecutive epochs → Training Duration: 8 epochs (early stopping typically triggered after 5-6 epochs)
Evaluation Metrics
→ Primary Metrics: Macro F1 score, Precision, Recall → Secondary Metrics: Accuracy → Evaluation Method: Hold-out test set (15% of data)

ArcaEfficacy (Efficacy Classification Model) - Performance Results - Overall Performance:		
Metric	Value	Notes
Macro Precision	93.52%	Ability to avoid false positives
Macro Recall	94.26%	Ability to find all sentences with benefits
Macro F1 Score	93.88%	Harmonic mean of precision and recall
Accuracy	96.72%	Overall correctness

ArcaEfficacy (Efficacy Classification Model) - Limitations and Considerations:	
Known Limitations	
→ The model may miss benefits expressed across multiple sentences → Lower performance on sentences with complex linguistic structures or extensive abbreviations	
Ethical Considerations and Biases	
→ Potential biases toward well-studied diseases and conventional treatment modalities → May underrepresent efficacy information for treatments studied primarily in non-English literature	

ArcaEfficacy (Efficacy Classification Model) - Maintenance and Support:	
Monitoring and Maintenance	
→ Monitoring Plan: Evaluation of newly published literature samples, every 3 months → Retraining Criteria: >3% decrease in F1 score on monitoring set or quarterly updates, regardless of performance	
Versioning	
→ Current Version: 1.1.3 → Update Frequency: Quarterly major revisions	

II.2.3.2. Model Card: Unified Relation Extraction Classification Model

Unified Relation Extraction Classification Model - Model Card			
Model Version:	1.0.0	Last Updated:	Jan 11, 2025
Type:	Encoder-based Text Classification	Language(s):	English
Framework:	PyTorch		

Unified Relation Extraction Classification Model - Model Description	
Overview	
This unified model evaluates sentences containing either efficacy information or risk mentions to determine if they specifically relate to the target or drug under investigation. It performs dual-task relation classification, analyzing contextual relationships between clinical outcomes (both positive and negative) and pathology mentions. The model categorizes relationships as "Relevant," "Possibly Relevant," or "Irrelevant" based on contextual analysis, serving as a critical component in both our efficacy and risk detection pipelines.	
Intended Use	
This model is intended to filter and classify both efficacy information and safety concerns specifically relevant to the investigation, enabling comprehensive evidence synthesis across the benefit-risk spectrum. It is designed for use in pharmaceutical research, clinical evidence assessment, pharmacovigilance, and therapeutic development pipelines where pathology-specific clinical outcome data is crucial for decision-making.	

Out-of-Scope Use

This model is not intended for:

- Complete relation extraction classification without upstream content detection components
- Processing non-biomedical text
- Direct clinical decision-making
- Evaluating outcome severity, frequency, or clinical significance

Unified Relation Extraction Classification Model - Model Architecture

Base Architecture:	pretrained biomedical model	Number of Parameters:	~340M
Input Format:	Tokenized text (maximum 512 tokens)	Output Format:	Three-class classification (0: Irrelevant, 1: Possibly relevant, 2: Relevant)

Key Architectural Features:

- Fine-tuned high-end healthtech encoder large model with multi-task learning capability
- Task-specific prefixes ([EFFICACY] and [RISK]) to indicate relation type
- Entity marker tokens to highlight clinical outcome, medication, and pathology spans
- Shared encoder with task-aware attention mechanisms
- Linear classification layer (1024 → 3) with softmax activation
- Task embedding layer to distinguish between efficacy and risk relation tasks

Unified Relation Extraction Classification Model - Training Details

Training Data

- Dataset(s): Combined proprietary corpus of biomedical literature with annotated efficacy-disease and risk-disease relations
- Total Size: 27,842 annotated sentences (9,842 efficacy relations + 18,000 risk relations)
- Data Format: Sentences with clinical outcomes and pathology mentions pre-identified, annotated with relation labels and task type
- Preprocessing: Sentence segmentation, entity marking with special tokens, task prefix addition
- Data Split:
 - ◆ 75% training,
 - ◆ 10% validation,

- ◆ 15% test (stratified by task type)
- Task Distribution:
 - ◆ Efficacy Relations: 35.3% of total dataset
 - ◆ Risk Relations: 64.7% of total dataset
- Class Distribution (Combined):
 - ◆ Relevant: 38.2%
 - ◆ Possibly Relevant: 13.1%
 - ◆ Irrelevant: 48.7%
- Class Distribution by Task:
 - ◆ Efficacy Relations: Relevant: 45%, Possibly Relevant: 20%, Irrelevant: 35%
 - ◆ Risk Relations: Relevant: 31%, Possibly Relevant: 6%, Irrelevant: 63%

Training Procedure

- Optimizer: AdamW
- Learning Rate: $2e-5$ with linear decay and warmup (1000 steps)
- Batch Size: 16 (balanced sampling across tasks)
- Sequence Length: 256
- Loss Function: Multi-task weighted focal loss with task balancing
- Task Weighting: Dynamic weighting based on task-specific performance
- Regularization: Dropout (0.1), L2 weight decay (0.01)
- Early Stopping Criteria: No improvement in combined validation F1 score for 4 consecutive epochs
- Training Duration: 12 epochs (early stopping typically triggered after 8-10 epochs)
- Multi-Task Training Strategy
 - ◆ Task Balancing: Alternating mini-batches between efficacy and risk tasks
 - ◆ Shared Representations: High-end Healthtech encoder for both tasks
 - ◆ Task-Specific Components: Task embedding layer and attention patterns
 - ◆ Loss Balancing: Weighted combination of task-specific focal losses ($\alpha=0.4$ for efficacy, $\alpha=0.6$ for risk)

Evaluation Metrics

- Primary Metrics: Task-specific and overall Macro F1 score, Precision, Recall
- Secondary Metrics: Accuracy, Task-specific confusion matrices
- Evaluation Method: Hold-out test set (15% of data) with task-stratified sampling

Unified Relation Extraction Classification Model - Performance Results - Overall Performance - Macro across both tasks:

Metric	Value	Notes
Overall Macro Precision	86.6%	Average performance across both tasks
Overall Macro Recall	85.3%	Average performance across both tasks
Overall Macro F1 Score	85.9%	Harmonic mean across tasks
Overall Accuracy	88.2%	Combined accuracy across all samples

Unified Relation Extraction Classification Model - Performance Results - Overall Performance - Efficacy Relation Classification:

Metric	Value	Notes
Macro Precision	89.4%	Consistent with standalone efficacy model
Macro Recall	86.2%	Maintained performance in unified architecture
Macro F1 Score	87.8%	No significant degradation from task consolidation
Accuracy	90.3%	Task-specific accuracy for efficacy relations

Unified Relation Extraction Classification Model - Performance Results - Overall Performance - Risk Relation Classification:

Metric	Value	Notes
Macro Precision	83.8%	Slight improvement over standalone model
Macro Recall	84.4%	Enhanced recall through shared representations

Unified Relation Extraction Classification Model - Performance Results - Overall Performance - <u>Risk Relation Classification</u> :		
Metric	Value	Notes
Macro F1 Score	84.1%	Improved F1 from multi-task learning
Accuracy	86.1%	Task-specific accuracy for risk relations

Unified Relation Extraction Classification Model - Cross-Task Analysis:
Task Interference Analysis: → Minimal negative transfer observed between tasks → Shared biomedical representations improve generalization → Task-specific attention patterns successfully maintained → Risk classification benefits from efficacy-related training data

Unified Relation Extraction Classification Model - Performance Results - Error Analysis:
Common error patterns include: → 4.2% of errors involve task misclassification (model treating efficacy as risk or vice versa)

Unified Relation Extraction Classification Model - Limitations and Considerations:
Known Limitations → Performance may decrease when efficacy and risk mentions appear simultaneously in complex sentences → May struggle with implicit pathology references or pronouns referring to previously mentioned conditions → Task identification relies on correct upstream classification (efficacy vs. risk detection) → Efficacy Relations: ◆ Performance varies with complexity of causal relationships described → Risk Relations: ◆ Lower recall for "Possibly Relevant" class due to inherent subjectivity
Ethical Considerations and Biases → Should not be used for direct clinical decision-making without human review

Unified Relation Extraction Classification Model - Maintenance and Support:
Monitoring and Maintenance

- Monitoring Plan: Quarterly evaluation of newly published literature samples for both tasks
- Performance Thresholds: >3% decrease in combined F1 score triggers review
- Task Balance Monitoring: Regular assessment of relative task performance to detect drift
- Retraining Criteria: Semi-annual updates regardless of performance, immediate retraining if threshold violations occur

Versioning

- Current Version: q.0.0
- Update Frequency: Semi-annual major revisions
- Backward Compatibility: Maintained through versioned API endpoints

Unified Relation Extraction Classification Model - Usage Guidelines:
Input Format Requirements:

- Sentences must include task prefix: [EFFICACY] or [RISK]
- Entity markers must properly identify investigation mentions
- Maximum input length: 256 tokens after tokenization

Integration Considerations:

- Model requires task type specification from upstream classification models
- Performance optimal when used with properly filtered and standardized input
- Recommended confidence thresholds: >0.85 for "Relevant", >0.65 for "Possibly Relevant"

II.2.4. Risk Detection Pipeline

Our risk detection pipeline identifies and categorizes safety concerns from biomedical literature through a sequential multi-stage process. First, our in-house safety classification model screens sentences to identify potential risk mentions, detecting both explicit statements and subtle indicators. This initial broad screening captures safety concerns regardless of severity or relation to specific treatments or conditions.

Following classification, we filter out sentences that mention our drug of interest while retaining all pathology-related content using our drug ontology. This approach prevents bias toward our specific intervention while capturing valuable risk relationships across related pathologies that might reveal additional therapeutic insights. We deliberately

maintain mentions of competitor drugs and standard-of-care treatments to establish comparative safety profiles and understand the broader therapeutic landscape within our target disease area.

Identified risk mentions are then standardized using the MedDRA terminology, mapping varied descriptions to consistent terms. Based on the identified MedDRA label, we can infer the System Organ Class. This standardization enables reliable categorization across studies and organizes risks at appropriate levels of specificity, from broad System Organ Classes to specific Preferred Terms.

Next, our **unified relation extraction (RE) classification model** analyzes each risk mention to determine its association with the investigation, distinguishing between risks inherent to the disease itself, risks from standard treatments, and unrelated findings. This contextual analysis is essential for accurately characterizing the risk profile specifically relevant to our investigation area. The model shows particularly strong performance in recognizing both risks and benefits described as direct consequences of the underlying intervention.

II.2.4.1. Model Card: ArcaPV(Safety Classification Model)

Model Card: ArcaPV(Safety Classification Model) - Model Card			
Model Version:	1.2.1	Last Updated:	May 20, 2025
Type:	Encoder-based Text Classification	Language(s):	English
Framework:	PyTorch	Created by:	Data Enrichment Team

Model Card: ArcaPV(Safety Classification Model) - Model Description
Overview
This model identifies sentences in biomedical articles that contain mentions of risks, adverse events, or safety concerns. It serves as the first component in our risk detection pipeline, performing initial screening of text to isolate safety-relevant content for further processing. The model has been trained to recognize diverse expressions of risk across various domains of biomedical literature.
Intended Use
This model is intended to filter large volumes of biomedical text to identify sentences containing risk-related information, which can then be processed by downstream components for standardization and relationship analysis. It is designed for use in pharmacovigilance, safety signal detection, and comprehensive risk profiling.

Out-of-Scope Use

This model is not intended for:

- Direct clinical decision-making
- Processing non-biomedical text
- Comprehensive risk assessment without downstream components
- Determining causality between treatments and adverse events
- Evaluating risk severity or frequency

Model Card: ArcaPV(Safety Classification Model) - Model Architecture

Base Architecture:	pretrained biomedical model	Number of Parameters:	~110M
Input Format:	Tokenized text (maximum 512 tokens)	Output Format:	Three-class classification (0: Irrelevant, 1: Possibly relevant, 2: Relevant)
Key Architectural Features:			
→ Fine-tuned high-end healthtech encoder large model with classification head → Linear layer (768 → 3) with softmax activation → Pretrained on biomedical corpus for domain-specific language understanding			

Model Card: ArcaPV(Safety Classification Model) - Training Details

Training Data

- Dataset(s): Proprietary annotated biomedical articles with risk mentions
- Size: 25,810 annotated sentences
- Data Format: Sentences extracted from biomedical articles with binary labels
- Preprocessing: Sentence segmentation, direct tokenization using high-end healthtech encoder tokenizer
- Data Split:
 - ◆ 70% training,
 - ◆ 15% validation,
 - ◆ 15% test
- For Classification:
 - ◆ Classes: Three-class (Relevant / Possibly relevant / Irrelevant)
 - ◆ Class Distribution: 28.1% relevant, 9.3% possibly relevant, 62.6% irrelevant

Training Procedure

- Optimizer: AdamW
- Learning Rate: 2e-5 with linear decay
- Batch Size: 16
- Sequence Length: 512
- Training Steps/Epochs: 4 epochs
- Loss Function: cross-entropy
- Regularization: Dropout (0.1)
- Early Stopping Criteria: No improvement in validation F1 score for 3 consecutive epochs

Evaluation Metrics

- Primary Metrics: F1 score, Precision, Recall
- Secondary Metrics: Accuracy
- Evaluation Method: Hold-out test set (15% of data)

Model Card: ArcaPV(Safety Classification Model) - Performance Results - Overall Performance:

Metric	Value	Notes
Precision	95.35%	Ability to avoid false positives
Recall	96.30%	Ability to find all risk mentions
F1 Score	95.81%	Harmonic mean of precision and recall
Accuracy	98.01%	Overall correctness

Model Card: ArcaPV (Safety Classification Model) - Limitations and Considerations:**Known Limitations**

- Performance varies based on how explicitly the risk is stated

Model Card: ArcaPV(Safety Classification Model) - Maintenance and Support:**Monitoring and Maintenance**

- Monitoring Plan: Evaluation on sample of new literature every 2 months
- Retraining Criteria: >3% decrease in F1 score on monitoring set

Technical & AI Models key findings

- ☐ The generation of the BRA dataset, the foundation of the BRA product, is a multi-step process where many AI models and treatments have been applied.
- ☐ The performance of specialized models is higher in biomedical tasks than that of general-purpose models.
- ☐ Many types of AI models and treatments have been used, such as named entity recognition (NER), relation extraction, large language models (LLMs), and classification models.

III. Quality process description

III. 1. Executive summary

This section presents a comprehensive validation of the AI models developed under the Benefit-Risk Assessment framework, focused on their readiness for deployment in pharmaceutical decision-making in accordance with the FDA's 2025 guidance on AI in regulatory contexts.

The BRA framework utilizes multiple AI techniques—including sentence classification, Named Entity Recognition, Relation extraction models, and Large Language Models—to extract, classify, and contextualize disease-target associations, safety risks, and therapeutic benefits from biomedical literature, supported by biomedical ontologies.

Model risk was comprehensively assessed by considering the influence of AI in the decision process and the significance of potential errors, resulting in overall risk levels from LOW-MEDIUM for regulatory support and benefit extraction to MEDIUM for early target identification and safety signal detection, with final decisions anchored by both AI output and expert judgment.

Internal validation demonstrated robust inter-annotator agreement and strong model performance: extraction tasks for molecular features and benefits consistently achieved F1 scores above 87%, with precision rates as high as 95%, though somewhat lower precision was noted in risk and benefit labeling (73–82%). When directly benchmarked against curated data sources such as DisGeNET, the BRA models showed strong concordance for top-ranked disease-gene associations and delivered extraction performance comparable to established AI systems in the biomedical domain.

To ensure ongoing quality and compliance, the BRA framework mandates regular performance monitoring, expert validation of AI outputs, manual fallback options, systematic ontology and literature updates, and alignment with evolving regulatory best practices.

In conclusion, while the BRA AI models consistently meet industry and regulatory standards for extraction and classification across most tasks, their adoption is firmly supported by continuous expert oversight, rigorous validation, and a robust risk mitigation strategy, ensuring that outputs are a reliable aid—rather than a stand-alone substitute—for expert-led benefit-risk assessments in drug development.

III. 2 .Aim of this section

This quality validation document provides a comprehensive evaluation of the AI models developed within the Benefit-Risk Assessment (BRA) framework, focusing on their ability to support informed decision-making for potential production deployment. It assesses whether the models meet the necessary performance standards for accurately extracting and classifying molecular mechanisms, safety risks, and therapeutic benefits from biomedical literature. This evaluation was conducted in alignment with the FDA's Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products: Guidance for Industry and Other Interested Parties, published in January 2025. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/consideration-s-use-artificial-intelligence-support-regulatory-decision-making-drug-and-biological>

III. 3 .Context of Use for the AI Models

The BRA framework employs multiple AI models including sentence classification models, Named Entity Recognition , and Large Language Models to extract information related to disease-target associations, risks, and benefits. The labeling process leverages LLMs in combination with biomedical ontologies.

III. 4 .The AI Model Risk

This risk assessment follows the FDA framework for AI/ML medical device risk evaluation, focusing on two critical dimensions: Model Influence (the contribution of AI-derived evidence relative to other evidence sources) and Decision Consequence (the significance of adverse outcomes from incorrect AI-based decisions).

Risk Categorization Matrix

Application Area	Model Influence	Decision Consequence	Risk Level
Early Target Identification	Low	Medium-High	Medium
Safety Signal Detection	Medium	Medium	Medium
Benefit Assessment	Medium	Low	Low-Medium
Regulatory Submission Support	Medium	Low	Low-Medium

III. 4 .1. Early Target Identification

Early Target Identification: Analysis
Model Influence Analysis
→ <u>Supporting Evidence:</u> Proprietary preclinical data, internal clinical data, expert scientific judgment, patent landscape analysis

- Influence Level: LOW - AI provides supplementary literature evidence but company relies primarily on internal data and expert assessment and other modelisation platforms
- Decision Integration: AI outputs serve as additional validation layer for decisions primarily driven by internal evidence

Decision Consequence Analysis

- Incorrect Target Prioritization (False Positive): Medium-High consequence
 - ◆ Major investment commitment
 - ◆ Resource allocation away from potentially better targets
 - ◆ Opportunity cost in competitive therapeutic areas
- Missed Target Opportunity (False Negative): Medium-High consequence
 - ◆ Competitive disadvantage if target proves successful for competitors
 - ◆ Lost first-mover advantage in therapeutic area

Overall Risk Classification: MEDIUM

III. 4 .2. Safety Signal Detection

Safety Signal Detection: Analysis

Model Influence Analysis

- Supporting Evidence: Internal clinical trial data, pharmacovigilance databases, regulatory guidance, expert clinical judgment
- Influence Level: MEDIUM - AI provides systematic literature screening but users exercise independent judgment on each signal

Decision Consequence Analysis

- False Positive Safety Signal: Medium consequence
 - ◆ Additional investigation time and resources
 - ◆ Enhanced monitoring in risk management plans
- Missed Safety Signal (False Negative): Medium consequence
 - ◆ Signals should be detected in clinical phases if clinically relevant
 - ◆ Primarily impacts comprehensive risk management planning

Overall Risk Classification: MEDIUM

III. 4 .3. Benefit Assessment

Benefit Assessment: Analysis	
Model Influence Analysis	
→ <u>Supporting Evidence</u> :	Internal clinical trial data, market research, clinical expert opinion
→ <u>Influence Level</u> :	MEDIUM - AI provides systematic competitive intelligence but is balanced with internal clinical evidence
Decision Consequence Analysis	
→ <u>Overestimated Benefits (False Positive)</u> :	Low consequence
◆	Primarily affects internal communications and competitive intelligence
→ <u>Missed Benefit Opportunities (False Negative)</u> :	Low consequence
◆	Limited impact on core clinical development strategy
◆	Primary benefit evidence comes from internal clinical trials
◆	Affects competitive analysis rather than product development
Overall Risk Classification: LOW-MEDIUM	

III. 4 .4. Regulatory Submission Support

Regulatory Submission Support: Analysis	
Model Influence Analysis	
→ <u>Supporting Evidence</u> :	Internal clinical and preclinical data, regulatory guidance, expert regulatory affairs judgment
→ <u>Influence Level</u> :	MEDIUM - AI provides systematic literature compilation but regulatory teams review and validate all outputs with authorities
Decision Consequence Analysis	
→ <u>Incorrect Literature Evidence (False Positive/Negative)</u> :	Low consequence
◆	Regulatory submissions reviewed extensively by internal teams
◆	Authorities evaluate all evidence sources independently
◆	Literature evidence is supplementary to core clinical data package
◆	Multiple review cycles allow correction of any literature inaccuracies
Overall Risk Classification: LOW-MEDIUM	

III. 5. AI Model Performance

III. 5. 1. Unitary Models Quality Validation

The performance of all models used in the BRA project has already been detailed in the previous section. The following validations aimed to confirm that each model is fit for purpose and trained on relevant and reliable data.

III. 5. 2. Dataset Quality Validation (within the context of use)

The performance of all models used in the BRA project has already been detailed in the previous section. The following validations aimed to confirm that each model is fit for purpose and trained on relevant and reliable data.

The BRA framework already executed has aimed to answer the question: *Do public data provide sufficient evidence to support the benefit-risk assessment of a drug in its early development stage?*

To answer whether public data can sufficiently support the benefit-risk assessment of a drug in its early development stage, several key capabilities are essential:

- Ability to relate diseases to biomarkers: Identifying and validating biomarker-disease associations to support mechanistic plausibility and predictive value in BRA.
- Ability to detect, standardize, and qualify risks across the drug development lifecycle—from early-stage compounds, by extracting and interpreting preclinical safety signals, target-related liabilities, and structural alerts in the absence of clinical data, to approved or marketed drugs, by leveraging real-world evidence (RWE), post-marketing surveillance, and clinical trial data to assess known adverse events and population-specific risks.
- Ability to detect, standardize, and qualify benefits across the drug development continuum—from early-stage development, by inferring potential benefits through target-disease alignment, drug class precedents, preclinical efficacy models, and translational insights in the absence of clinical data, to approved indications, by analyzing clinical trial outcomes and real-world effectiveness to contextualize benefits within established use cases.
- Ability to extract contexts in which benefits and risks occur: Capturing drug modalities (e.g., biologics, small molecules), dosing regimens, and relevant patient characteristics (e.g., age, comorbidities, sex) to define the conditions under which benefits and risks manifest.

To address these questions, the framework has been divided into four steps, which will be detailed in the following sections. The process is based on the following annotation sheet: <https://docs.google.com/spreadsheets/d/1md7um0u0tPAH7WGSwAp15yvZLx0PL7P00ceTr4Qfld8/edit?gid=0#gid=0>

III.5.2.1. Disease-Associated Target Identification

This step involves applying a sentence classification model, supported by multiple ontologies and a large language model for relation labeling.

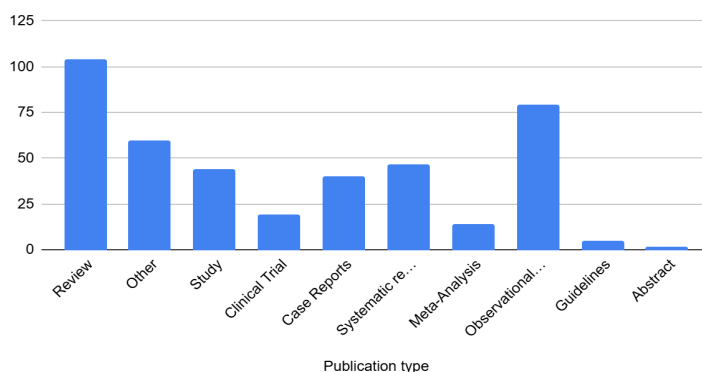
The aim of this validation step is to ensure that the model—trained on a diverse range of subjects and treatment contexts—is capable of accurately extracting relevant information for the intended use case: the detection of molecular features in Hidradenitis Suppurativa. This phase focuses exclusively on the extraction task, which forms the foundation of the pipeline. If the extraction demonstrates sufficient precision and reliability based on predefined performance metrics (e.g., precision, recall, F1-score), it can be considered a successful and trustworthy component.

This extraction step serves as the input for the subsequent labeling phase, where biomedical ontologies are used to standardize the names of genes, proteins, and biomarkers. Given the controlled vocabulary used in this step, the risk of hallucination is minimal. Finally, a large language model is employed to classify the nature of the molecular relations (e.g., upregulation, downregulation, altered, non-altered, unknown). Since the range of expected outputs is limited and well-defined, the likelihood of hallucination remains low.

Internal Validation - Extraction task: Creation of the Golden standard	
Source	
The full Hidradenitis Suppurativa dataset, including both positive and negative articles (i.e., with and without molecular mechanism extractions), was compiled from PubMed and Medline without restricting by publication timeframe.	
Sample Size Justification	
To estimate the performance of our AI model in detecting molecular mechanisms within the HS literature, we refer to established practices in biomedical natural language processing (NLP) for determining adequate sample sizes: → In a recent study on clinical named entity recognition (NER), Hu et al. (2024) annotated 254 documents (163 positive, 91 negative) to validate LLM performance [JAMIA, 2024, doi: 10.1093/jamia/ocad259]. → The Journal of Biomedical Informatics recommends a gold standard corpus of approximately 500 annotated sentences for NLP evaluation tasks [doi: 10.1016/j.jbi.2011.12.010]. → A methodological study on sample size for fine-tuning large language models in biomedical NER also identified a point of diminishing returns at 440–530 sentences [Majdik et al.]. → Since our evaluation is based on article-level sampling—selecting entire articles rather than individual sentences—while still validating each sentence within those articles, we chose this approach for feasibility. In total, we annotated 228 selected articles related to Hidradenitis Suppurativa. Among them, 138 contained at least one sentence positively annotated by the model, while 90 contained no positively annotated sentences (purposive sampling).	
Representativeness of the Sample	
The selected validation sample was intentionally heterogeneous and representative of the overall dataset in the following dimensions: → <u>Sources</u>: 84 articles (37%) from PMC and 144 (63%) from medline	

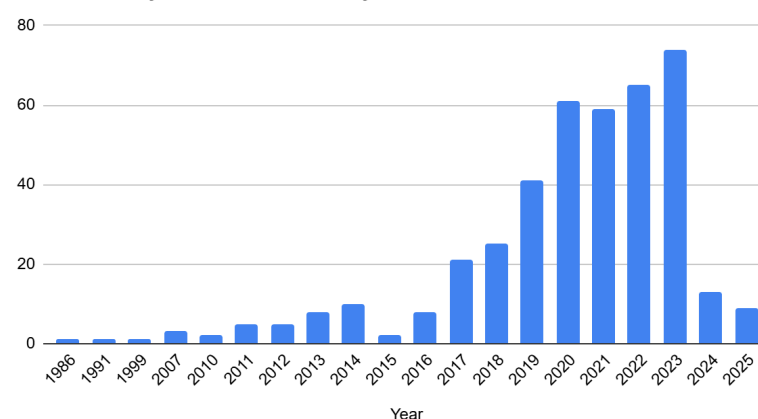
- Publication type: As shown in the graph, all publication types were included, reflecting the true distribution of the initial dataset.

Publication Type distribution by Article Count



- Publication Date: As shown in the graph, the included articles span from 1986 to 2025, taking into account the year distribution of the initial dataset.

Publication year distribution by article count



Annotation Team

The manual validation of molecular mechanism extractions was conducted by:

- C.A., Doctor of Pharmacy (PharmD) and holder of a Master 2 in preclinical and clinical Pharmacology (M2R)
- R.K., Doctor of Pharmacy (PharmD) and holder of a Master 2 in toxicology (M2R)

Annotation Guidelines Summary (Not the full guideline)

- Annotators read the full text excluding tables, figures, and their captions.
- Extract sentences that describe a relation between the pathology “HS” and any of the following entities: biomarkers, genes, sRNAs, hormones, cells, immune markers, including cases where no relation is present.
- Sentences may or may not include quantitative metrics.
- Exclude sentences that:
 - ◆ Are not about HS.
 - ◆ Describe relations between two pathologies.

- ◆ Appear in Materials and Methods sections indicating that the relation will be studied (future tense)
- If the same information appears multiple times in the article, not extracting repeated mentions is tolerated (i.e., missing repeats is acceptable).
- Extraction should cover the full sentence.
- If a compound sentence contains relevant information, it should be labeled as positive.
- Ambiguous or borderline cases: Annotators should use their best judgment; disagreements are reviewed and resolved during synchronous meetings.
- Formatting: Sentences should be preserved exactly as in the original text.
- Negations or contradictory statements: Sentences negating or contradicting a relation should still be extracted and counted as relevant.
- Annotation files: Annotations are recorded independently by each annotator and shared for cross-reference.

Internal Validation - Extraction task: Annotation Results

Inter-Annotator Agreement

Inter-annotator agreement was assessed using Cohen's Kappa, yielding a high score of 0.82, which indicates almost perfect agreement between annotators. To resolve remaining disagreements, the annotators engaged in an adjudication process, discussing conflicting annotations synchronously in meetings to reach consensus and ensure consistency.

Extraction Metrics

To evaluate extraction performance, we computed precision, recall, and F1-score based on manual validation of model-predicted positives and a sample of predicted negatives. As true negatives were not fully annotated, accuracy, specificity, and AUC-ROC could not be reliably assessed.

Calculated metrics were:

Precision	93.05%
Recall	87.31%
F1	90.05%

These results validate the ability of the overall product—comprising the extraction model and its supporting components—to effectively fulfill its intended purpose: identifying relevant molecular features associated with Hidradenitis Suppurativa.

The high precision (93.05%) confirms the reliability of extracted information, while the strong recall (87.31%) indicates substantial coverage of relevant content.

The resulting F1-score (90.05%) demonstrates a robust balance between relevance and completeness.

External Validation

We used DisGeNET as an external validation source for our work. To enable a meaningful comparison between our results and those from DisGeNET, we noted that DisGeNET lists a total of 250 genes or entities associated with Hidradenitis Suppurativa, while our approach identified 675 entities. We conducted a one-to-one comparison of the 10 highest-ranked entities in DisGeNET to determine whether they are also present in the Arcascience dataset. The results of this comparison are presented in the following table.

Entity	Found in AS
TNF	Yes
NCSTN	Yes
PSENEN	Yes
MEFV	Yes
PSEN1	Yes
PSTPIP1	Yes
NOD2	Yes
GJB2	Yes
IL17A	Yes
CRP	Yes

III.5.2.2. Risk Extraction

This step involves applying a sentence classification model to determine whether a sentence relates to a safety concern, with support from multiple ontologies, primarily MeSH. A large language model is used only as a support tool to ensure that the models and treatments employed are appropriate for the context of use—namely, the extraction and labeling of preclinical and clinical safety risks. The aim of this validation step is to confirm that our pipeline can accurately and reliably extract risk-related information from articles and correctly label it.

Internal Validation - Extraction task: Creation of the Golden standard

Source

The gold standard was generated from the initial dataset related to the OX40/OX40L axis (query terms: OX40, OX40L, their synonyms, and related medications), as well as from articles referencing adalimumab. The selection of articles and sentences was purposely conducted to address the significant imbalance between positive and negative sentences, while remaining random with respect to all other factors such as publication date, article type, and query term (Annotators do not know about the model's results when annotating).

Sample Size Justification

As mentioned previously, validation of NER models typically involves around 500 sentences or approximately 250 articles. Since this is a first implementation and our dataset is heterogeneous—focusing on adalimumab and OX40/OX40L—we decided to exceed this usual amount. We employed purposive sampling to achieve balanced representation of positive and negative examples, yielding a dataset containing 805 non-relevant and 481 relevant sentences. This sampling approach produced a gold standard dataset of 560 articles: 270 articles with no model-annotated positive sentences and 290 articles containing at least one positively annotated sentence.

Representativeness of the Sample

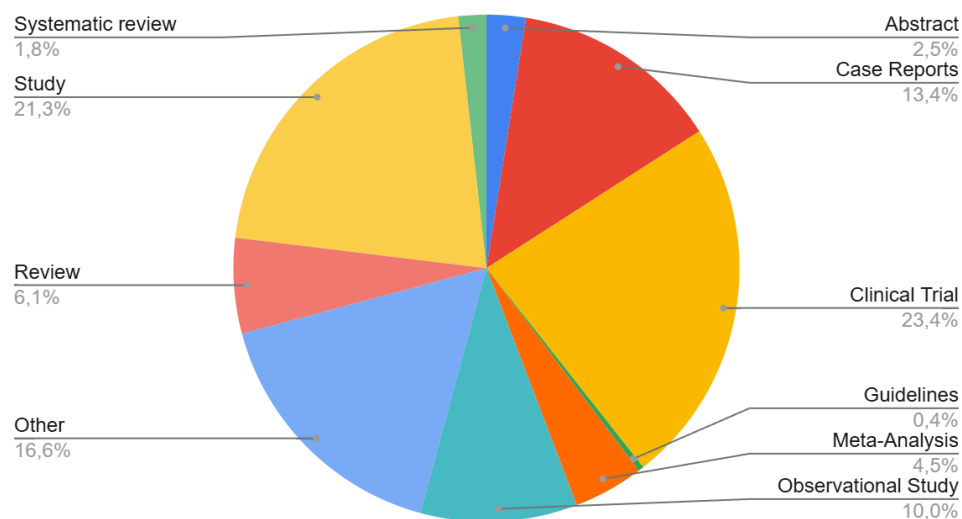
The selected validation sample was intentionally heterogeneous and representative of the overall dataset in the following dimensions:

→ Source:

The sample of articles used in this study was sourced from two major databases: Medline and PMC. Out of the total dataset, 249 articles were retrieved from Medline, while 311 articles originated from PMC.

→ Publication type:

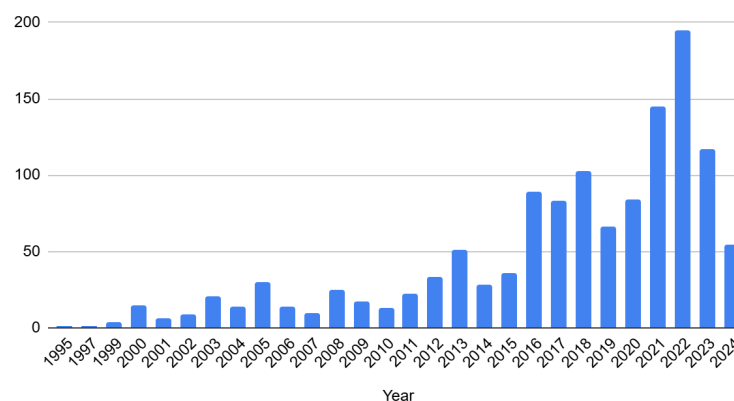
Distribution of Articles by Publication Type



The evidence used in the annotation set comes from various publication types, with a high proportion of case reports, observational studies, and clinical trials.

→ Publication Date:

Distribution of annotated sentences by publication date

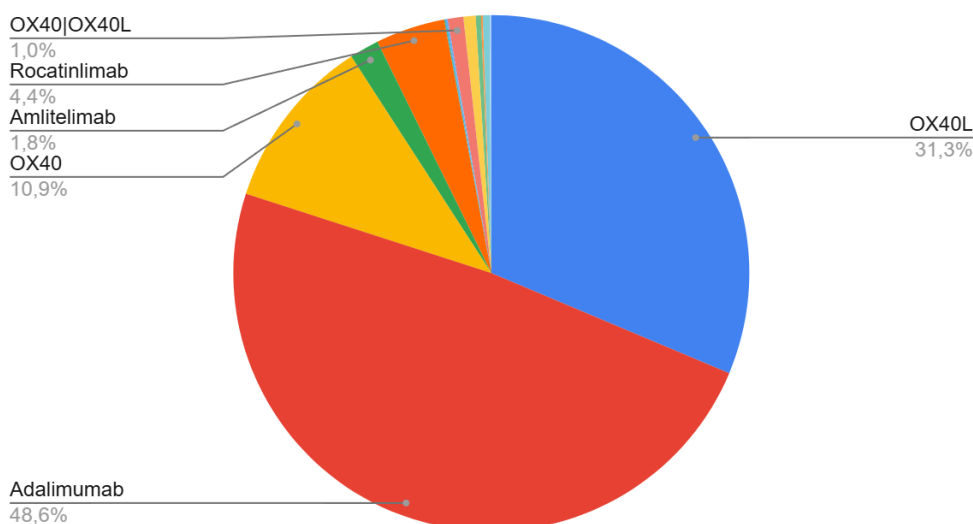


The figure below shows that the manually annotated sentences come from articles published between 1995 and 2024, reflecting the overall publication trend during these years.

→ Query term:

The sample dataset was generated for both adalimumab and OX40/OX40L. The distribution of these terms across the sentences in the dataset is illustrated in the figure below.

Query Term Distribution Across the Sample



Annotation Team

The manual validation of risk extractions was conducted by:

- C.A., Doctor of Pharmacy (PharmD) and holder of a Master 2 in preclinical and clinical Pharmacology (M2R)
- R.K., Doctor of Pharmacy (PharmD) and holder of a Master 2 in toxicology (M2R)

Annotation Guidelines Summary

Objective

- The goal of this annotation task is to evaluate whether each sentence from selected articles is relevant or irrelevant for identifying risk information related to the inhibition of OX40/OX40L axis or to adalimumab. This will support the validation of our pipeline's ability to accurately extract and label risk-related content.

Input for Annotation

- Annotators receive all sentences from selected articles that include at least one predefined search term. Annotators are also provided with the paragraph context from which each sentence was extracted.

Annotation Task

- Annotation Guidelines

Your task: Label each sentence as either Relevant or Irrelevant based on whether it describes risks related to OX40/OX40L interference or adalimumab.

- ◆ Label as RELEVANT when a sentence mentions risks associated with:
 - OX40/OX40L disruption: Any negative effects from blocking, reducing, altering, or genetic variations in OX40 or OX40L

- OX40/OX40L-targeting therapies: Risks from any drug or treatment that works by targeting these pathways
- Adalimumab (Humira): Any risk mentioned in connection with this medication or its biosimilars
- ◆ Important notes for Relevant labeling:
 - Include sentences about risk even when causality isn't proven or directly stated
 - Accept risk descriptions at any level (lab studies, animal models, human studies)
 - Quantitative data (percentages, ratios) isn't required—qualitative risk descriptions count too
- ◆ Label as IRRELEVANT when sentences describe:
 - Benefits or positive effects of OX40/OX40L interference or adalimumab
 - Disease relationships that don't involve OX40/OX40L or adalimumab risks
 - Unrelated risks not connected to these specific pathways or medications
 - Methods sections or future research plans (sentences using future tense like "will be assessed")
 - Lack of efficacy (ineffectiveness rather than safety risks)
- ◆ **Key principle: Focus specifically on safety risks and adverse effects related to interfering with the OX40/OX40L system or using adalimumab—everything else should be labeled Irrelevant.**
- Special Cases
 - ◆ If the same information appears multiple times in an article, repeated mentions do not need to be re-annotated. It is acceptable to miss these duplicates. If a case is ambiguous or borderline, annotators should use their best judgment. All disagreements or uncertainties is resolved in synchronous review meetings.
 - ◆ Sentences that negate or contradict a risk association should still be annotated as relevant. These sentences contribute to the understanding of the risk landscape.
- Labeling Standards
 - ◆ All risk terms must be labeled using MedDRA (Medical Dictionary for Regulatory Activities) terminology. If an exact MedDRA term is not available, annotators should choose the closest matching term from the MedDRA hierarchy.
- Annotation Format
 - ◆ Annotators must preserve the sentence wording exactly as it appears in the original article. Annotators should record annotations independently. Annotated files must be shared for cross-reference and discussion.

Internal Validation - Extraction task: Annotation Results

Inter-Annotator Agreement

Inter-annotator agreement was assessed using Cohen's Kappa coefficient on 1,286 annotations, yielding a high score of $\kappa = 0.894$, which indicates almost perfect agreement between annotators according to Landis and Koch's guidelines ($\kappa > 0.80$). The annotators achieved 95.0% raw agreement, disagreeing on only 64 cases (5.0%).

To resolve remaining disagreements, the annotators engaged in a structured adjudication process, discussing conflicting annotations synchronously in meetings to reach consensus.

Extraction Metrics

The model demonstrates strong performance in identifying relevant cases, with a recall of 88.9%, meaning it successfully captures the vast majority of true positives. However, the precision is moderately lower at 73.4%, indicating the presence of some false positives. This results in an F1 score of 80.5%, reflecting a solid balance between precision and recall. Notably, the drop in precision is largely due to the inclusion of the “lack of efficacy” label, which alone accounts for 62 false positive cases. When this label is excluded, the adjusted precision improves significantly to 84.3%, and the F1 score increases to 86.5%, demonstrating a more favorable precision-recall trade-off. No overall accuracy was calculated, as the dataset is based on purposive sampling, making precision and recall more appropriate for evaluating model performance.

Metric	Original Value	Adjusted Value
Precision	73.40%	84.1%
Recall	88.89%	88.97%
F1 Score	80.46%	86.4%
False Positives	128	66

These results validate the ability of the overall product to effectively fulfill its intended purpose: identifying relevant molecular features associated with Hidradenitis Suppurativa.

Labeling Metrics

The labeling process combined the use of the MedDRA dictionary with a large language model (LLM) to identify terms not found in the dictionary. A large portion of false positives stemmed from sentences incorrectly predicted as relevant by previous models, especially those related to the “lack of efficacy” label. After adjusting for these cases, the precision improved significantly to about 72%, reflecting a more accurate identification of relevant sentences.

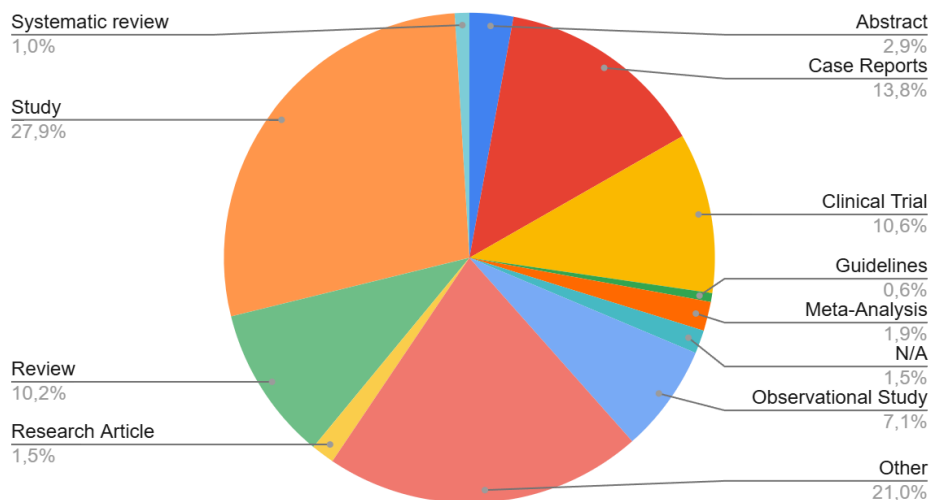
Metric	Original Value	Adjusted Value (±95%CI)
Precision	59.0%	72.0%
Recall	86.6%	86.6%
F1 Score	70.1%	78.1%

III.5.2.3. Benefit Extraction

This step involves applying a sentence classification model to determine whether a sentence relates to a positive outcome, with support from multiple ontologies—primarily MeSH and an internally developed COA ontology. A large language model is used solely as a support tool to ensure that the selected models and treatments are appropriate for the intended context of use, namely the extraction and labeling of preclinical and clinical benefits. The objective of this validation step is to confirm that our pipeline can accurately and reliably extract benefit-related information from articles and assign correct labels.

Internal Validation - Extraction task: Creation of the Golden standard	
Source	
The gold standard was generated from the initial dataset related to the OX40/OX40L axis (using query terms such as OX40, OX40L, their synonyms, and related medications), as well as from articles referencing adalimumab in the context of hidradenitis suppurativa. The selection of articles and sentences was intentionally designed to address the significant imbalance between positive and negative sentences, while remaining random with respect to all other factors, including publication date, article type, and query terms.	
Sample Size Justification	
The sample size used for benefit extraction validation is comparable to that used for risk validation: a total of 874 sentences were assessed. Purposive sampling was employed for the same reasons explained in the risk validation section. Specifically, 538 irrelevant sentences (from 373 articles) and 337 relevant sentences (from 189 articles) were evaluated. Annotators were blinded to the model's results during the annotation process.	
Representativeness of the Sample	
The selected validation sample was intentionally heterogeneous and representative of the overall dataset in the following dimensions: → <u>Source</u> : The sample of articles used in this study was sourced from two major databases: Medline and PMC. Out of the total dataset, 132 articles were retrieved from Medline, while 430 articles originated from PMC. → <u>Publication type</u> :	

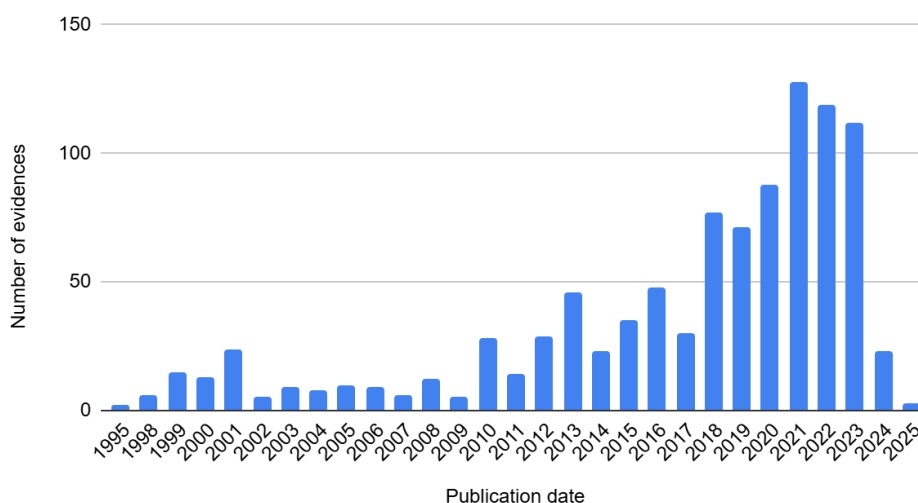
Distribution of Articles by Publication Type



All publication types were included in the sample dataset, with the highest proportion coming from studies, reviews, case reports, and clinical trials.

→ Publication Date:

Distribution of annotated sentences by Publication Date

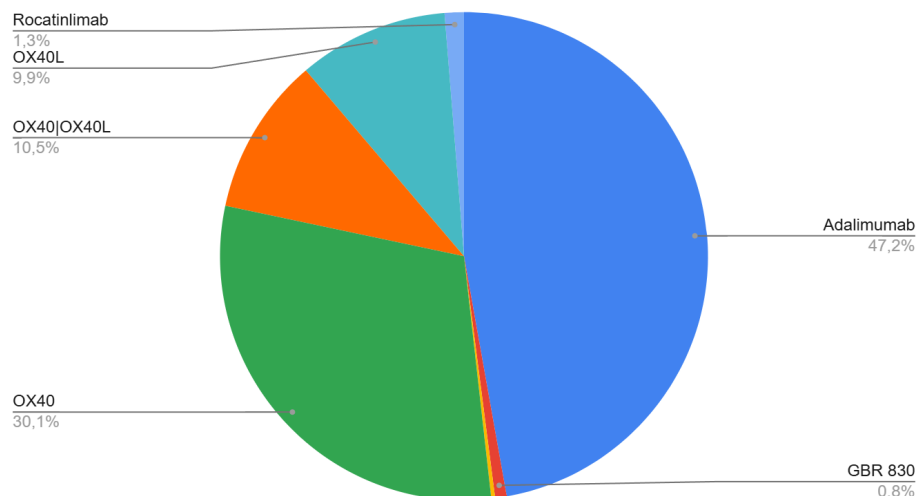


Manually annotated sentences come from publications dating from 1995 to 2025, with a high proportion originating from the past seven years, which represents the peak of publication activity on this topic.

→ Query term:

The sample dataset was generated for both adalimumab and OX40/OX40L. The distribution of these terms across the sentences in the dataset is illustrated in the figure below.

Query term distribution across the sample



Annotation Team

The manual validation of the benefit extractions was conducted by:

- C.A., Doctor of Pharmacy (PharmD) and holder of a Master 2 in preclinical and clinical Pharmacology (M2R)
- R.K., Doctor of Pharmacy (PharmD) and holder of a Master 2 in toxicology (M2R)

Annotation Guidelines Summary

→ Objective

You'll evaluate sentences to identify benefit information related to blocking the OX40/OX40L pathway or using adalimumab, specifically in the context of Hidradenitis Suppurativa (HS) or HS-related conditions.

→ What You'll Work With

- ◆ Sentences containing predefined search terms from selected articles
- ◆ Paragraph context for each sentence to help with interpretation

→ Labeling Instructions

→ Label as RELEVANT when a sentence describes benefits from:

- ◆ OX40/OX40L interference: Blocking, reducing, altering, or genetic variations in these pathways
- ◆ OX40/OX40L-targeting treatments: Any therapy that works by targeting these pathways
- ◆ Adalimumab (Humira): Including biosimilar versions

→ A sentence is RELEVANT if it mentions benefits in any of these contexts:

- ◆ Direct HS benefits: Improvements specifically in Hidradenitis Suppurativa patients
- ◆ HS-related molecular pathways: Benefits from modulating pathways involved in HS (like Th17 inhibition)
- ◆ Shared symptoms: Improvements in symptoms common to both HS and other conditions (e.g., reduced itching in atopic dermatitis, since HS patients also experience itching)

- ◆ Quantitative data optional: Benefits with or without specific numbers/percentages
- Label as IRRELEVANT when sentences describe:
 - ◆ Risks or side effects from OX40/OX40L blocking or adalimumab use
 - ◆ Unrelated benefits not connected to OX40/OX40L or adalimumab, or connected but not relevant to HS
 - ◆ Methods sections or future research (using future tense like "will be assessed")
 - ◆ Molecular changes with no clear connection to HS improvement
- Special Considerations
 - ◆ Repeated information: Don't worry about re-annotating identical content—missing duplicates is acceptable
 - ◆ Unclear cases: Use your best judgment; uncertainties will be discussed in team meetings
 - ◆ Contradictory statements: Even sentences that contradict or negate benefits should be labeled as relevant, as they still inform our understanding
- Benefit Labeling Standards
 When you identify a benefit, label it using:
 - ◆ Official COA tool names when applicable (e.g., "Dermatology Life Quality Index (DLQI)")
 - ◆ Standardized terms for other benefits
 - ◆ Full biomarker names when benefits involve biomarkers
 - ◆ Standardized direction phrases like "Improvement in..." or "Decrease in..."
 Example: "Improvement in Dermatology Life Quality Index (DLQI)" or "Decrease in serum C-reactive protein levels"
- Annotation Process
 - ◆ Keep original sentence wording exactly as written
 - ◆ Work independently, then share files for team review and discussion

Internal Validation - Extraction task: Annotation results

Inter-Annotator Agreement

Inter-annotator agreement was assessed using Cohen's Kappa coefficient on 875 annotations, yielding a high score of $\kappa = 0.84$. To resolve remaining disagreements, the annotators engaged in a structured adjudication process, discussing conflicting annotations synchronously in meetings to reach consensus.

Extraction Metrics

While the extraction of benefits task achieved a strong recall of 94%, indicating that most actual benefit sentences were identified, the model's precision was 0.819 (95% CI: 0.784–0.854). This means that although the model is effective at finding benefits, it might also incorrectly classifies certain sentences as benefits when they are not. The F1 score, combining both metrics, was 0.875 (95% CI: 0.853–0.897) falls within expected norms.

Metric	Value
Precision	0.819
Recall	0.941
F1 Score	0.875

Labeling Metrics

Our benefit identification evaluation shows strong recall (0.925), successfully capturing nearly all actual benefits. Precision is moderate (0.683), with most benefit labels correctly identified. The precision gap stems primarily from preclinical benefit labeling complexity, while clinical labeling achieved higher accuracy. These results provide a solid foundation with clear opportunities for optimizing preclinical identification.

Metric	Value
Recall	0.925
Precision	0.683
F1-score	0.786

External validation of the benefit and risk

While there is no published data that mimics the job that has been done for the OX40/OX40L axis or adalimumab benefits and risks in HS, we cannot perform a direct comparison. Many papers have been published about the use of AI in literature reviewing, and the results demonstrate reasonably strong extraction performance, comparable to state-of-the-art models such as UniNER-7B (84.8) and GLiNER-L (82.9) for their F1 scores in **general medical applications** [Zaratiana et al., 2023]. However, regarding large language models (LLMs), Scherbakov et al. [2025] highlighted inconsistent performance: while data extraction and abstract screening with LLMs show variable but overall moderate-to-good results (typically between 60% and 90% when outliers are excluded), performance remains poor and unreliable for searching publications, with major fluctuations and numerous outliers.

These findings echo those of recent systematic reviews, which have shown that conversational LLMs like ChatGPT perform well for general summarization and knowledge queries in healthcare, but also reveal significant limitations regarding reliability, accuracy, and the ability to manage complex clinical questions [Wang et al., 2024]. Editorial work on open-source models such as DeepSeek emphasizes that, although these models provide benefits in terms of cost, adaptability, and privacy (due to possible offline deployment), they still pose concerns over privacy, bias, and output reliability, necessitating rigorous human oversight and regulation before clinical application [Temsah et al., 2025]. Furthermore, practical evaluations have demonstrated that LLMs can generate critical errors and propagate hallucinations in clinical data tasks—including tumor staging—with

error rates and fabricated responses that may compromise scientific validity unless carefully reviewed [Kwong et al., 2024].

Overall, despite the growing promise of LLMs and AI for scientific literature review, current evidence suggests these tools are not yet mature enough to operate independently—especially for tasks such as reference identification, relevant data extraction, and ensuring consistency and homogenization across multiple queries. Human expertise, methodological rigor, and strong regulatory frameworks remain essential to ensure results are accurate, trustworthy, and ethically sound—particularly for complex, specialized, or high-impact clinical topics.

III.5.2.4. Context Identification.

This step focuses on identifying information related to how drugs have been used, the types of articles retrieved (e.g., clinical trials, case reports), and the patient characteristics in which the reported benefits or risks occurred. In summary, it aims to define the clinical context in which these benefits and risks were observed.

Since this task is not specific to the topic of the OX40/OX40L axis in Hidradenitis Suppurativa, and does not directly impact the model's performance, its results can be reasonably extrapolated from the overall model performance. Moreover, as this step is not central to the core functionality of the product, it does not require a separate performance assessment tied to this particular use case.

III.5.2.5. Small-Scale Aesthetic Corrections

In addition to the AI-related issues, small fixes need to be made to the final product. There are occasional duplicates at the sentence level, where the same sentence may appear with an additional part. Some extracted sentences start with punctuation marks from the previous sentence—such as a bracket, comma, or even a period. Others begin with fragments like “Fig...”. While these extraction issues do not affect the outcome of the AI process, they are important for the aesthetic quality of the product. The extent of the problem is limited—duplicates occur in approximately 1 to 3% of cases, and punctuation issues affect no more than 2%.

III.5.2.6. Methodology Limitations

The methodology adopted for validation introduced several notable limitations. First, while purposive sampling was employed to mitigate the issue of class imbalance, this strategy reduces the generalizability of findings, as the selected samples may not fully reflect the distribution of entities in real-world datasets and can introduce selection bias. The scope of validation was further restricted to sentence-level extraction and labeling accuracy; this granularity, though practical, may fail to capture contextual cues and associations present across broader sections of documents, potentially overlooking subtle errors or nuanced relationships that only emerge at the document level (rare case). Although the sample size was considered adequate in comparison to similar NER validation efforts, the quality and consistency of annotation heavily relied on the expertise of curators with specialized backgrounds in pharmacology and toxicology. This reliance introduces the possibility of interpretive variability, particularly between clinical and preclinical contexts, and the adjudication process may have also contributed to unintentional group bias.

III. 6 . Risk Mitigation Strategy

ArcaScience's risk management approach addresses two distinct categories of operational risks that could impact service delivery and data quality.

III. 6 .1. SaaS Infrastructure Risk Management

Risks associated with the Software as a Service (SaaS) environment include system availability, security vulnerabilities, data loss, server maintenance issues, and technical infrastructure failures that may impact service continuity. To mitigate these risks, we have implemented a set of protocols covering server redundancy, security controls, automated backup procedures, and disaster recovery strategies.

Our global IT security strategy is defined in the IT Security Insurance Plan, which outlines the security policies, operational standards, and governance practices in place to protect systems and data. Similarly, our IT Quality Insurance Plan describes the processes and methodologies we apply to ensure performance, reliability, and service quality across all IT deliverables.

In support of continuous improvement and industry best practices, our processes are aligned with recognized frameworks and standards, including those relevant to ISO 27001. This alignment is part of our preparation for formal ISO 27001 homologation, scheduled to begin at the start of next year, which will further strengthen our information security management system (ISMS) and reinforce our commitment to compliance and risk management.

III. 6 .2. Data Exhaustiveness Monitoring

Our data exhaustiveness risk management relies on automated monitoring systems that ensure comprehensive coverage of source materials. We conduct weekly automated tests that systematically compare our data repository against original sources including PubMed and clinical trial databases, with alert systems that trigger immediate notifications for any significant deviation in data coverage. Additionally, our daily document update system operates under supervised automatic tests using Airflow workflows, providing continuous oversight of data ingestion processes and immediate detection of any gaps or inconsistencies.

III. 6 .3. AI Model Performance Evaluation

AI model performance is continuously monitored through regular evaluation cycles that occur every 2 to 6 months depending on the specific model type and usage frequency. Our performance threshold monitoring system automatically flags any F1-score decrease of 5% or greater, triggering immediate investigation protocols to identify and address performance degradation. To maintain optimal performance over time, we implement annual model retraining using updated datasets and validation protocols, ensuring our models remain current with the latest scientific literature and maintain their predictive accuracy.

III. 6 .4. High-Risk Task Quality Control

For sensitive data or high-risk analysis tasks, we implement prospective quality assurance measures that involve conducting quality checks on 15 to 35% of data before client delivery. This process includes comprehensive performance evaluation and transparent communication of results to clients, allowing them to understand the confidence levels associated with their specific analysis. When performance metrics fall below acceptable thresholds during these pre-delivery assessments, we immediately initiate model re-evaluation and correction procedures, ensuring clients receive only validated, high-quality results.

III. 6 .5. User Feedback Integration

Our user feedback integration system creates a continuous improvement loop that captures real-world performance insights after delivery. The frequency of feedback collection is customized based on individual client requirements and project criticality levels, with systematic analysis of user-reported issues and performance concerns feeding directly into our quality improvement processes. When feedback indicates performance issues, we implement immediate investigation and corrective measures, as demonstrated in cases like a client project remediation detailed in the annex. This feedback is systematically integrated into future model training and validation cycles, ensuring continuous enhancement of our analytical capabilities based on actual user experience and requirements.

Quality processes key findings

- ☐ The BRA framework achieves F1 scores above 85% for extraction tasks with 100% concordance against DisGeNET for top disease-gene associations, meeting industry standards for pharmaceutical AI applications.
- ☐ Risk assessment categorizes applications from LOW-MEDIUM to MEDIUM levels.
- ☐ While extraction performance meets standards, preclinical benefit labeling precision requires enhancement
- ☐ The validation aligns with the FDA's 2025 AI guidance, incorporating question definition, context of AI use, risk assessment, AI performance within the intended context of use, and a corresponding risk mitigation strategy.

IV. Technology deployment outcomes

Executive summary

To build on the recent implementation process and deliver transformational value, we propose deploying a computational platform that will help Sanofi teams conduct Benefit-Risk Assessments and literature reviews across its organization. This initiative will migrate crucial data analysis tasks from siloed, project-based activities to integrated, organization-wide operational capabilities.

- Key Objectives and Solution Highlights:

The solution delivers tailored dashboards and analytical tools for each user role, ensures consistent and well-referenced data, and securely integrates public and proprietary Sanofi sources to enhance BRA insights. By shifting to a dynamic web-based platform with real-time data and API integration, it streamlines collaboration and enables continuous, proactive risk monitoring and faster decision-making across the organization.

- Platform Capabilities:

The platform features highly customizable widget-based dashboards, powerful GenAI-driven data exploration tools for full transparency and traceability, and automated literature monitoring with personalized alerts. Multi-channel data integration ensures continuous, comprehensive surveillance for regulatory and pharmacovigilance, with planned extensions to include social media data.

- Business Impact and ROI:

Deployment of the platform streamlines the entire pharmaceutical value chain at Sanofi, from target validation to regulatory submissions and pharmacovigilance, delivering up to 71% time savings on literature review tasks and annual cost reductions exceeding €100,000, with an estimated ROI of 27%. By reducing project delays, reliance on external consultants, and enabling earlier risk identification, the platform empowers more agile, data-driven decisions and accelerates time-to-market across the organization.

- Deployment Approach:

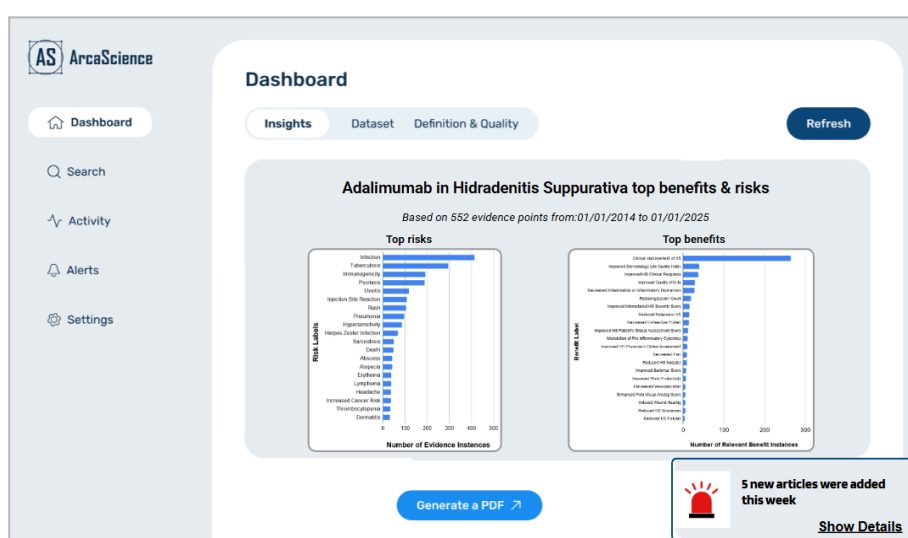
A robust deployment plan will maximize value by customizing dashboards for user needs, securely integrating diverse data sources, and ensuring effective adoption through targeted communication and training.

Conclusion:

This deployment will equip Sanofi's teams with a scalable, secure, and continuously evolving platform—simplifying benefit-risk assessment, accelerating drug development, and ensuring regulatory compliance.

IV.1. Deployment objectives

Building on our latest implementation results, we propose deploying a comprehensive computational platform that will transform how Sanofi teams conduct Benefit Risk Assessments and literature reviews at scale. This deployment will shift these critical activities from isolated, project-specific tasks to integrated, daily operational capabilities accessible across the organization.



Example of Data insight on ArcaScience platform

The deployment will deliver five core capabilities that will simplify and enhance Sanofi's approach to benefit-risk monitoring:

IV.1.1. Deploy Customized User Interfaces for Specific Use Cases

We implement diverse visualization solutions that adapt to the specific needs of each user profile. Building on our implementation deliverables, each stakeholder benefits from a selection of widgets aligned with their role, while retaining full access to the underlying data for independent exploration.

IV.1.2. Implement Advanced Data Structuring and Referential Integrity

This technology empowers any Sanofi team member to conduct instant literature reviews and access structured, insightful, and well-referenced datasets. The platform ensures data consistency and traceability across all analyses.

IV.1.3. Augment and Diversify Data Sources

Beyond the public sources demonstrated in the implementation, our platform can securely integrate proprietary literature accessible via API, confidential Sanofi documents, and real-time data feeds from multiple sources. When deployed on Sanofi infrastructure, dedicated pipelines will ensure the highest levels of security and confidentiality while significantly expanding analytical capabilities.

IV.1.4. Enable Organization-Wide Data Accessibility

We will transition from static CSV file delivery to a dynamic, web-based platform approach that offers all users direct access, seamless API integration with existing business applications, and the elimination of technical configuration barriers. This shift enables real-time data synchronization across teams, ensuring faster, more collaborative, and more efficient decision-making.

IV.1.5. Transform BRA from Project-Based to Continuous Operations

Traditional CRO models operate on a per-project basis, resulting in siloed data that is reviewed once and used for a single purpose. In contrast, our integrated Benefit-Risk Assessment platform provides continuous access to evolving datasets, enabling real-time benefit-risk monitoring, proactive risk identification, enhanced decision-making capabilities, and immediate access to updated analyses for any team member.

This will empower teams to enhance decision-making and monitor the evolution of benefit-risk evaluation in real-time, thereby anticipating risks and opportunities as early as possible.

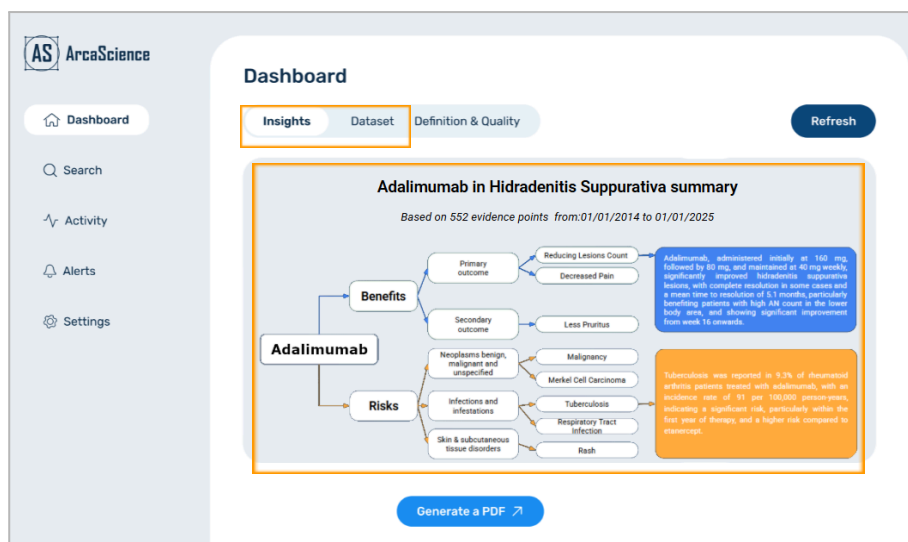
IV.2. Solution description

We propose deploying a comprehensive computational platform that will transform how Sanofi teams conduct Benefit Risk Assessments and literature reviews at scale. This SaaS platform will deliver the following key characteristics:

IV.2.1. Customizable Visualization Capabilities

IV.2.1.1. Dynamic Dashboarding

Users can select their preferred visualization types, variables, and timeframes to create personalized analytical views. The platform offers multiple configurable options, including BRA decision trees, risk-benefit heatmaps, statistical histograms, and trend analysis charts. All visualizations are fully customizable and provide direct links to source documents, ensuring complete traceability from insight to original data.



In orange menu & widgets can be customized

IV.2.1.2. Widget Types

To prevent information overload, the platform employs a widget-based architecture tailored to each user persona. Each widget can be customized with specific parameters, allowing users to focus on their most relevant analyses without being overwhelmed by unnecessary options.

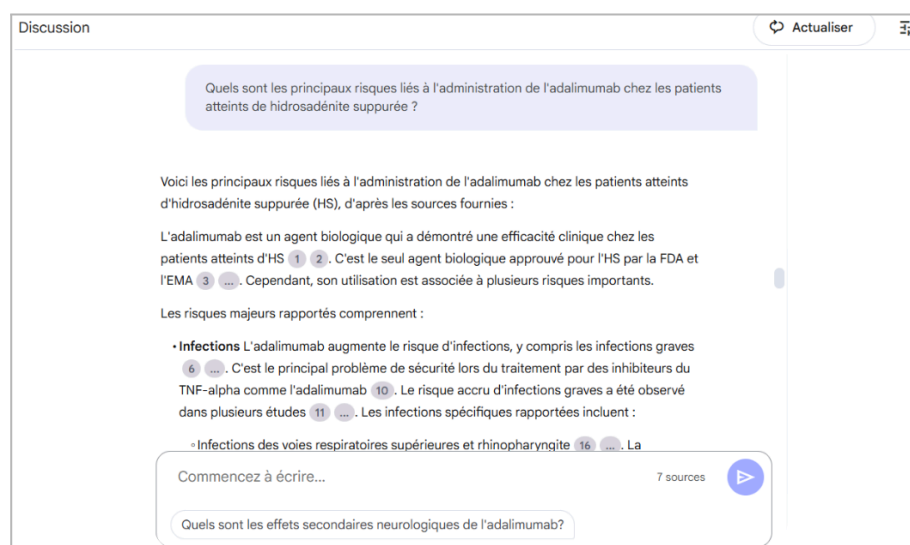


Multiple widgets availables

IV.2.1.3. Data exploration tools

The platform's drill-down capability marks a significant advancement over traditional CRO deliverables—typically static reports with limited traceability—and commercial large language models. It enables users to verify 100% of references and extracted data

through integrated chatbot functionality for guided exploration, direct access to underlying tabular data, and interactive visualizations that link back to source materials. This ensures complete transparency and validation of all analytical outputs.



Gen AI available to explore the dataset

IV.2.2. Data Integration and Sources

By default, the platform connects to a set of public data sources, including PubMed, ClinicalTrials.gov, PMC, and other. This foundation ensures immediate access to high-quality, publicly available information. Initial platform configuration is then customized for each client to define additional relevant data sources and access methods. Proprietary and institutional data—such as internal research datasets, clinical trial results, hospital records, and confidential Sanofi documents—can be securely integrated. Subscription-based resources like Scopus, Web of Science, and specialized archives are accessed via automated authentication systems that comply with licensing and user subscription protocols. The platform also incorporates social media and patient language analysis through advanced NLP models trained on real-world patient discourse, enabling detection of adverse events, sentiment analysis, and real-world evidence mining (planned for Q1 2026).



Integration of various sources is possible, subject to the terms of Sanofi's existing contracts.

IV.2.3. Monitoring (Alert System) Functionality

IV.2.3.1 Personalized Alert Configuration:

Users can define custom monitoring parameters such as specific therapeutic targets or drug compounds, disease areas and pathological conditions. The system then automatically detects relevant publications and extracts qualified information, enabling intelligent filtering and prioritization based on the user's predefined criteria.

IV.2.3.2 Notification Frequency and Format:

Users maintain complete control over alert frequency and delivery, with customizable notification schedules ranging from daily to weekly, or monthly updates. A dedicated alert section within the platform interface centralizes all notifications, which are structured to highlight PICOS components—Population, Intervention, Comparator, Outcomes, and Study design—of detected BRA findings. Each alert delivers contextualized information to support rapid assessment and informed decision-making.

IV.2.3. Preliminary financial approach

Price per pathology	Budget in € without taxes
Upfront Costs (Year 1)	
ArcaScience solution deployment for 5 profiles	98 000
Additional public data source set-up for each data sources	11 200
Additional subscription based data source set-up for each data sources	22 500
Private data sources	pricing on demands
Total Upfront Costs Included 5 profiles + 1 additional public source + 1 additional subscription based sources	131 700 €
Ongoing Costs (Year n)	
ArcaScience solution Subscription Fees	85 000
Datasource Maintenances (60% of Up front cost)	20 220
Total Ongoing Costs Included 5 profiles + 1 additional public source + 1 additional subscription based sources	105 220 €

This estimation will be adjusted based on Sanofi's expectations and condition targeted, which will be investigated before the order is signed.

IV.2. Use Case and ROI

After this comprehensive explanation of model performance, underlying technology principles, and the types of information this system can extract and analyze, it's crucial to address the fundamental question: how does this technology translate into practical value for Sanofi professionals in their daily work? What are the concrete applications that can transform routine processes and decision-making?

First, it's important to clarify that what has been demonstrated in this proof of concept represents the raw analytical capabilities of the system—essentially showing how the data extraction and classification functions perform. However, the final commercial product would be delivered through user-friendly interfaces including interactive dashboards, conversational chat interfaces, and comprehensive data tables, all supported by relevant scientific references and citations (as mentioned before). This ensures that users receive actionable insights seamlessly integrated into their existing workflows, rather than having to interpret raw model outputs.

The BRA framework's versatility enables numerous practical applications across the pharmaceutical value chain. Many compelling use cases emerge from this foundational technology, each addressing specific pain points and operational challenges faced by different teams within pharmaceutical organizations. These applications span from early

discovery through post-market surveillance, offering opportunities to accelerate and consolidate decision-making and reduce costs (time and externalisation costs). Processes that previously took weeks or months can now be completed in a matter of hours or days.

The implementation of these use cases demonstrate a 27% return on investment.

Use case	Time (hours)	Project delay (month)	Cost saved (€uro)	Non-quantifiable benefits
Target validation and deeper understanding of disease biology	500	4	28k	→ Accelerated Decision-Making → Reduced R&D Waste through Optimized Competitive Intelligence
Regulatory Submission Acceleration	786	6,5	32k	→ Accelerated Time to Market → Process Transformation
Pharmacovigilance Enhancement	1 584	NA	48k	→ Proactive Risk Management
TOTAL benefits	2 870	10,5	108k	The ROI of the solution is 27.06%
TOTAL solution ongoing cost (with no additional data source)			85k	

The following are details of potential use cases (non-exhaustive) for the BRA product at Sanofi with ROI descriptions:

Target validation and deeper understanding of disease biology
Description
Target validation emerges as a key use case where AI-driven literature analysis can significantly enhance early-stage drug discovery. By enabling rapid, systematic assessment of disease–target associations, this approach supports a deeper understanding of disease biology and therapeutic mechanisms. Leveraging AI-powered extraction and classification of global scientific publications, Sanofi teams can efficiently compare multiple therapeutic targets or approaches in parallel, accelerating go/no-go decisions. This use case helps identify potential safety liabilities earlier, reduces time spent on manual literature review, and ultimately guides R&D efforts toward the most promising and evidence-backed opportunities.
Strategic ROI
→ <u>Accelerated Decision-Making:</u> Reduces the time to evaluate and validate a therapeutic target from

approximately 12 weeks to just 3 weeks, enabling faster portfolio progression. More in details, McKinsey reports that process improvements and automation in early development—driven by digital and analytics—can cut cycle time from candidate nomination to first-in-human by $\approx 40\%$.

[\[https://www.mckinsey.com/industries/life-sciences/our-insights/operational-excellence-in-biopharma-research-and-early-development?utm_source=chatgpt.com\]](https://www.mckinsey.com/industries/life-sciences/our-insights/operational-excellence-in-biopharma-research-and-early-development?utm_source=chatgpt.com)

→ Reduced R&D Waste through Optimized Competitive Intelligence:

Prevents late-stage failures by flagging unviable targets early, thereby reducing unnecessary preclinical investments through improved visibility on both scientific and competitive evidence. More specifically, industry estimates suggest that applying digital and automation strategies across R&D could yield up to \$1.2–\$1.7 billion in savings per approved medicine (Accenture, 2020).

Regulatory Submission Acceleration

Description

This use case supports teams across the drug cycle development in preparing literature-based components of regulatory submissions. In preclinical, teams must assemble documents such as Investigator's Brochures (IBs), Target Product Profiles (TPPs), and nonclinical overviews—tasks that require thorough literature support. Later, during clinical development and market approval stages, submissions to regulatory authorities (e.g., IND, CTA, NDA, MAA) require comparisons with existing and approved therapeutic approaches. Automating the literature review process helps reduce the time and manual effort typically involved, improves consistency, and enables teams to build more complete, evidence-based submissions more efficiently.

Strategic ROI

→ Accelerated Time to Market

By reducing literature preparation time by up to 60–80%, this approach enables faster regulatory submissions and earlier access to the market, providing a competitive advantage.

→ Process Transformation

Shifts regulatory documentation from a one-time, static assessment to a continuous, real-time evaluation, improving responsiveness and adaptability throughout the drug development lifecycle.

Pharmacovigilance Enhancement
Description Continuous monitoring of safety signals across scientific literature enables earlier identification of risks and supports proactive risk management strategies. This use case integrates seamlessly with BRA and pharmacovigilance teams by delivering relevant safety data regularly—daily or weekly—eliminating time spent manually searching databases.
Strategic ROI → <u>Proactive Risk Management:</u> Enables earlier detection of safety issues, reducing the likelihood of costly late-stage safety failures and supporting safer clinical development.

Future Deployment – Social Media–Based BRA Monitoring
Description A future opportunity within the BRA framework involves expanding monitoring capabilities to include social media platforms, patient advocacy sites, and digital health communities. This extension would enable earlier detection of rare adverse events, uncover patient-reported outcomes, and offer real-world insights into drug effectiveness—complementing traditional scientific literature. While not yet implemented, this use case is under active development.
Strategic ROI → Expanding into social media and forum monitoring can enable detection of up to 38% more safety signals, 3–6 months earlier than traditional systems, while uncovering real-world patient insights and unmet needs often missed by conventional sources. This information can also support drug repurposing by identifying off-label uses and emerging treatment patterns directly from patients [KOSS, et al, 2021] [https://regconsult.co/regulatory-affairs-news/social-listening-for-pharmacovigilance-in-europe-where-are-we-in-2021/?utm_source=chatgpt.com]

In addition to the strategic returns on investment mentioned above, financial return on investment can also be observed. The financial ROI analysis is based on three primary value drivers: time savings, through reduced full-time equivalent hours spent on literature review and analysis tasks; cost avoidance, by decreasing reliance on external consultants and contract research organizations; and risk mitigation, enabled by earlier identification of issues that prevent costly late-stage failures. The estimates provided below are based on user interviews with experts working in benefit-risk assessment, pharmacovigilance,

and regulatory affairs. While these figures are grounded in real-world experience, they may not fully reflect Sanofi's internal benchmarks.

Query Parameters	Resource Costs
• 2000 articles to be screened initially • Working day: 7 hours • Literature screening efficiency: 25 articles takes 2 hours • Article analysis time: 2 hours per relevant article (if determined relevant)	• Research Associate: €350/day (7 hours) • Senior Scientist: €800/day (7 hours) • Medical Writer: €400/day (7 hours)

Task	Human Resources	Time (Human)	Delay (Human)	Cost of Externalization
Target validation and deeper understanding of disease biology				
Literature Screening	Research Associate	160 hours (2000 articles ÷ 25 × 2h)	1,5 months	
Data Extraction & Standardization	Research Associate	200 hours (100 relevant articles × 2h)	1,5 months	
Data Analysis	Research Associate + Senior Scientist	112 hours (56h each)	0,5 month	
Report Generation	Medical Writer /Research Associate	28 hours (4 days)	0,5 month	
TOTAL	Multiple roles	500 hours	4 months	€25,000-40,000

ROI per project

- Time Savings: 557 hours (72.9% reduction)
- Internal Cost Savings: €28,825
- Cost Avoidance from CRO: €40,000
- Project Delay time: 4 months with no real time after the first delivery

Task	Human Resources	Time (Human)	Delay (Human)	Cost of Externalization
Regulatory Submission Acceleration				
Literature Screening	Research Associate	240 hours (3000 articles ÷ 25 × 2h)	2 months	
Data Extraction & Standardization	Research Associate	300 hours (150 relevant articles × 2h)	2,5 months	
Data Analysis	Research Associate + Senior Scientist	168 hours (84h each)	1,5 months	
Report Generation	Medical Writer /Research Associate	78 hours (11 days)	0,5 month	
TOTAL	Multiple roles	786 hours	6,6 months	€90,000-123,000

ROI per project

- Time Savings: 562 hours (71.5% reduction)
- Internal Cost Savings: €44,983 - €12,700 = €32,283
- Cost Avoidance from CRO: €90,000-123,000
- Project Delay time: 6,5 months with no real time after the first delivery

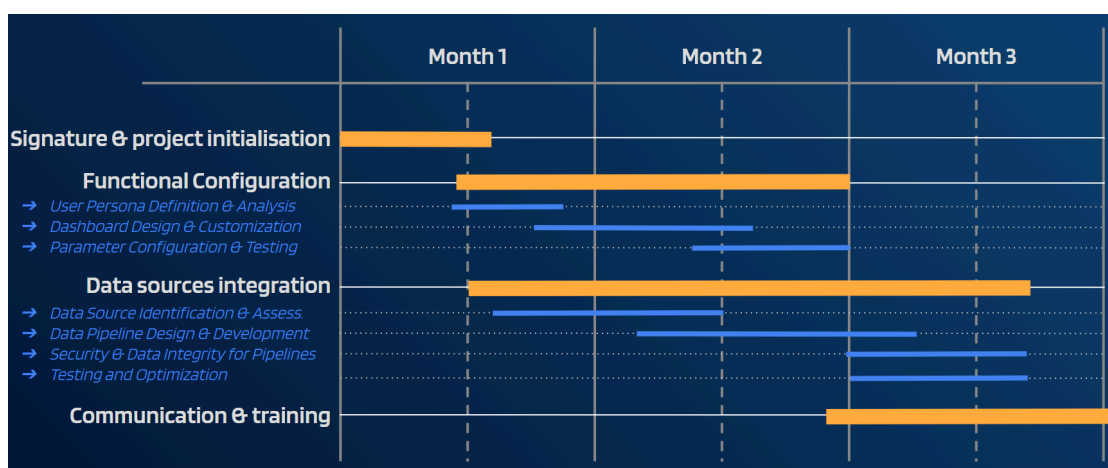
Task	Human Resources	Time (Human)	Time (with AI)	Cost of Externalization
Pharmacovigilance Enhancement (PER MONTH) ~10 marketed drugs (Mixte)				
Literature Screening	Research Associate	40 hours (500 articles ÷ 25 × 2h)	4 hours	
Data Extraction & Standardization	Research Associate	50 hours (25 relevant articles × 2h)	2.5 hours	
Data Analysis	Research Associate + Senior Scientist	28 hours (14h each)	21 hours (10.5h each)	
Report Generation	Medical Writer /Research Associate	14 hours (2 days)	10.5 hours (1.5 days)	
TOTAL	Multiple roles	132 hours/month	38 hours/month	€12,500-19,000/month
ROI per month: → <u>Time Savings:</u> 94 hours (71.2% reduction) → <u>Internal Cost Savings:</u> €7,580 - €2,675 = €4,905 → <u>Cost Avoidance from CRO:</u> €9,825-16,325/month				

Beyond the direct time and cost savings described above, the BRA AI system offers significant risk mitigation benefits that depend on the specific use case. The most quantifiable impact is seen in target validation. Selecting the wrong drug target — which often results in development failure — can increase the total cost of developing a new drug by approximately 30–40%. For example, the estimated average cost per approved drug rises from around \$780 million to \$1.2 billion (95% CI, \$888.1 million–\$1,480.8 million) when failures are factored in (Wouters et al., 2020). Such failures are common, as only 13.8% of therapeutic agents entering phase I ultimately receive FDA approval; the success rates are 21.0% for those starting phase II, and 59.0% for those in phase III (Wong, Siah, & Lo, 2019). For pharmacovigilance, early safety signal detection can prevent costly market withdrawals through proactive risk management, though the impact is difficult to quantify due to variability across therapeutic areas and patient populations. Regulatory submissions offer variable financial impact depending on drug type and market.

IV.3. Deployment approach

To successfully deploy this solution and maximize its impact for Sanofi, our team is implementing a robust project methodology. This project addresses three key dimensions critical to its success:

- Functional Configuration: renewing the SOW with new parameters
- Selection of Additional Document Sources: Identifying and integrating new document sources, along with assessing their impact on the technical architecture.
- Tool Training and Communication: Providing comprehensive training and clear communication about the new tool to all users.



Project timeframe to be confirmed for signature

IV.3.1 Functional Configuration - Profil parameters

This section focuses on, following the SOW update, activating ArcaScience solution's dashboards and functionalities to fit the specific needs and roles of different user profiles within Sanofi. In doing so, we maximize relevance and efficiency for each user.

1 - User Persona Definition & Analysis:

Identify Key User Roles:	Work with Sanofi stakeholders (e.g., R&D scientists, medical affairs, legal, compliance, business intelligence analysts, management) to set-up distinct user roles and their specific data consumption and interaction needs.
Conduct User Workshops & Interviews:	Gather detailed requirements for each operator, including the types of information they need to see, the metrics that are critical to their work, the desired level of granularity, and their typical workflows.
Map Existing Workflows:	Understand how users currently access and utilize data to identify pain points and opportunities for improvement with the new solution.

2 - Dashboard Customization:

Develop Profile-Specific Views:

For each identified operating user, select a widget to customize dashboard views, focusing on displaying only the most relevant information and functionalities. This might involve:

- **Personalized Widgets/Panels:** Allowing users to access tailored dashboard layouts with preferred widgets, charts, and data visualizations.
- **Filtering and Sorting Presets:** Creating pre-defined filters and sorting options relevant to specific roles (e.g., filter by therapeutic area, molecule, publication type).
- **Search and Alert Configurations:** adjusting search parameters and alert notifications to specific profile interests (e.g., alerts for new publications on a specific drug target for R&D scientists).

3 - Parameter Configuration & Testing:

System Configuration:

Configure the SaaS solution's backend to support the defined profile parameters, including setting up roles, permissions.

User Acceptance Testing (UAT) for Profiles:

Conduct thorough UAT with a diverse group of users from each profile to validate that the dashboards are intuitive, accurate, and meet their specific requirements. Document feedback and iterate on improvements.

Documentation of Profile Settings:

Create comprehensive documentation outlining the purpose and configuration of each dashboard parameter and profile setting.

IV.3.2 Data sources integration

This section details the critical process of integrating new data sources into the ArcaScience solution, including creating robust data pipelines, especially considering the hybrid nature of subscription-based literature and private Sanofi data. It is important to understand that private data & subscription based literature require to set-up a specific architecture. In the case of integration of public sources, the solution remains a full SaaS solution.

1 - Data Source Identification & Assessment:

Inventory of Data Sources:	Create a comprehensive list of all potential data sources, including: → <u>Subscription-Based Literature</u>: Databases like Embase, Web of Science, Scopus, industry-specific journals, etc. → <u>Internal Sanofi Private Data</u>: Clinical trial data, R&D reports, internal research documents, proprietary databases, competitive intelligence, etc. (requiring on-premise or secure cloud integration).
Data Governance and Compliance Review:	For each source, assess data quality, relevance, volume, refresh frequency and regulatory compliance (e.g., GDPR, HIPAA, internal Sanofi data security policies).
Technical Feasibility Assessment:	Evaluate the technical feasibility of integrating each source, considering APIs, data formats, security protocols, and potential data transformation needs.

2 - Data Pipeline Design & Development:

Architecture selection	If necessary, design a secure and scalable data architecture. For Sanofi's private data, this involves: → <u>On-Premise Data Connectors/Agents</u>: Deploying secure connectors within Sanofi's IT environment to extract private data. → <u>Cloud-Native Integration for SaaS Literature</u>: Utilizing the SaaS platform's native connectors or APIs for seamless integration with external literature databases.
AI model adaptation/parameters	For each new source selected, parameter all AI models applied in the project to be compatible with each new source.
Automated Pipeline Orchestration:	Set-up automation tools to schedule, monitor, and manage data ingestion pipelines, ensuring data freshness and reliability.

3 - Security & Data Integrity for Pipelines:

End-to-End Encryption:	Ensure data is encrypted in transit and at rest, especially for sensitive Sanofi private data.
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Access Control:	Implement strict access controls for data pipelines, limiting who can access and modify them.
Data Validation and Error Handling:	Build robust validation checks and error handling mechanisms within pipelines to identify and address data quality issues promptly.
Auditing and Logging:	Implement comprehensive logging and auditing of all data movements and transformations for compliance and troubleshooting.

4 - Testing and Optimization:

Unit and Integration Testing:	Thoroughly test each component of the data pipeline and the integration points between different sources and the SaaS.
Performance Testing:	Assess the performance of data pipelines under load to ensure they can handle current and future data volumes.
Data Reconciliation:	Verify data accuracy and completeness by reconciling data from source systems with the data in the SaaS solution.

IV.3.3 Communication & training

This section focuses on ensuring that Sanofi operators are well-informed, engaged, and proficient in using ArcaScience solutions.

1 - Communication Strategy Development:

Stakeholder Identification:	Identify all internal stakeholders (end-users, managers, IT, leadership) who need to be informed.
Key Message Definition:	Craft clear and concise messages about the "why," "what," and "how" of the new solution, emphasizing the benefits to users and Sanofi.
Communication Plan:	Develop a plan for communicating regularly through various channels: → <u>Official Announcements</u>: Formal emails from leadership, intranet announcements. → <u>Interview/Roadshows/Presentations</u>: Introductory sessions to showcase the solution's capabilities. → <u>Newsletters/Updates</u>: Regular updates on progress, new features, and tips.

2 - Training Program Design & Delivery:

Training Needs Assessment:	Identify all internal stakeholders (end-users, managers, IT, leadership) who need to be informed.
Customized Training Materials:	Adapt comprehensive and user-friendly training resources: → User Manuals/Guides: Step-by-step instructions for core functionalities. → FAQ: large FAQ tools with semester enrichment → Training session support: Predefine agenda with activities and training environment (if non SaaS solution deployed)
Training Rollout:	<u>Role-Based Training Sessions</u> : Conduct specific training sessions tailored to the dashboards and functionalities relevant to each user profile. These can be in-person workshops, virtual instructor-led training (VILT).

Quality processes key findings

- ☐ The Platform aims to transform isolated BRA tasks into integrated organization-wide capabilities.
- ☐ The platform offers customized interfaces, advanced data structuring, diverse source integration, and continuous operations.
- ☐ Solution addresses target validation, regulatory submission acceleration, and pharmacovigilance enhancement with strong ROI.
- ☐ Implementation involves functional configuration, secure data integration, and comprehensive user training programs.

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Annex: Example of Risk Management Strategy Application for Client X

Aim of the Project: The primary aim of this project was to leverage public scientific literature to systematically detect and extract documented risks and benefits associated with hemostatic devices in the field of Urology. This effort was designed to address a clear regulatory requirement for comprehensive, evidence-based benefit–risk assessments to support product safety and efficacy claims.

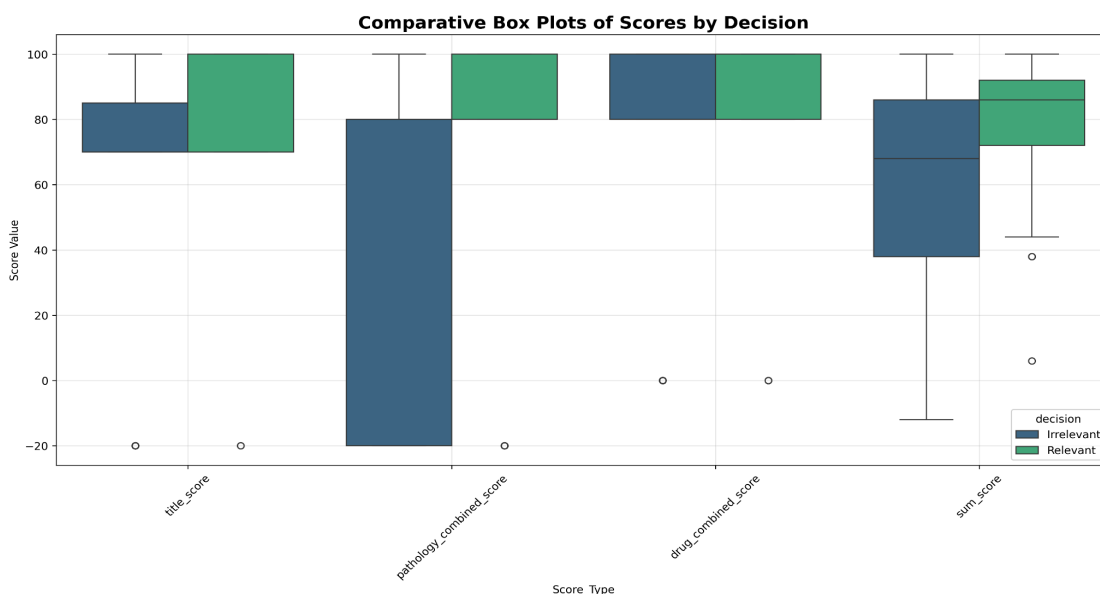
Modality of Client Interaction: As part of the risk mitigation and quality assurance process, all results delivered to the client were subsequently debriefed in a dedicated review session. This session provided an opportunity for the client to comment on the validation outcomes and clarify data interpretation needs. In addition to this interactive discussion, a formal quality control document was generated, summarizing performance metrics, data characteristics, and key findings, and serving as a record for both transparency and traceability.

Client Issue: While the client expressed overall satisfaction with the initial dataset—particularly regarding risk extraction—challenges arose due to the substantial volume of data related to benefits. The client indicated that, for regulatory submission and decision-making, it was impractical to review all extracted benefit statements and thus prioritized the need for tools to identify which items warranted focused scrutiny.

Proposed Solution: Contextual Relevance Scoring System To address this challenge, a contextual relevance scoring system was introduced. This system was designed to rank benefit statements according to their anticipated importance and contextual relevance for the client's regulatory use-case, thus prioritizing the information most likely to be meaningful in decision-making.

Results: The scoring system integrates three main criteria: Title Relevance, Disease Relevance, Hemostatic Device Relevance.

Performance and Outcome Statistical validation of the scoring system showed strong discriminative ability: Both Standard and Welch T-Tests demonstrated significant



differences in score distributions between statements marked as “Relevant” versus “Irrelevant” ($p < 0.0000001$, Cohen’s $d \approx 0.69$). The system proved effective in surfacing the most relevant content: 50% of relevant statements were placed in the document’s top segment (compared to only 5% in the bottom), while 92.5% of irrelevant statements were filtered to the bottom. This enhanced the client’s ability to quickly identify the benefit evidence most pertinent to regulatory needs, while minimizing time spent on less consequential information.

Conclusion: This approach demonstrated practical value as a risk mitigation tool: It improved the efficiency of regulatory review and documentation by focusing attention on high-priority evidence, while also establishing a transparent, data-driven method for benefit–risk documentation from the literature.